

The Cubic Boundary for Restricted Partitions

Sharp volume laws, uniform all-order expansions,
exponentially small corrections, inverse asymptotics,
and a Poisson law for repeated parts

Research report prepared for Vladimir Reshetnikov

1 October 2026

Abstract

Let $p_m(N)$ count partitions of N into parts at most m , equivalently partitions into at most m parts. We give an elementary necessary and sufficient condition for the volume approximation:

$$p_m(N) \sim \frac{N^{m-1}}{m!(m-1)!} \iff \frac{N}{m^3} \longrightarrow \infty, \quad m \longrightarrow \infty.$$

This proves, and sharpens, a condition still described as conjectural in OEIS A238016 and A258670. The sufficiency has classical antecedents and is not claimed as a new discovery. At the critical scale $N \sim cm^3$, the ratio to volume tends to $e^{1/(4c)}$. We prove a centered expansion uniform for $N \geq c_0m^3$, give an all-order coefficient algorithm and five explicit corrections for A238608, and derive a second expansion that resolves actual exponential sectors for $N = q^m$. Both types are inverted, with careful separation of smooth asymptotic charts from integer-valued inverse functions. In the conjugate model with at most m parts, the number of equal adjacent parts converges to $\text{Poisson}(1/(2c))$; its probability generating function has an explicit first finite-size correction. Exact computation checks 258 samples, including all displayed initial A238608 values, and the package supplies symbolic derivation and numerical verification code.

Status. This is a self-contained proof-and-extension report. Its leading partition formulas and first-wave machinery have substantial classical precedents. The explicit refinements below are derived here, but no exhaustive priority claim is made. The proofs have not been independently peer reviewed or formalized in Lean. Exact finite checks corroborate formulas; they do not replace the uniform estimates proved in the text.

Editorial note (5 October 2026, batch 98). Since batch 98 the report has two Parts. Part I is the report described above, manuscript 05 of batch 74, unchanged except for dated notes. Part II (§12) adds manuscript 09 of batch 98: one logarithmic expansion uniform for every $N \geq am^2$, the sharp centered threshold $X/m^{5/2} \rightarrow \infty$, a hierarchy of transition scales $m^{2+1/(2r)}$ with infinitely many nonzero constants, and an analytic inverse chart. It answers Part I's research question 2 and proves Theorem 3.1, the near-arc bound of Lemma 3.2, Corollary 4.1 and the correction $-61/288$ again (Table 2). Its labels carry the prefix `rpc:cx:`.

Contents

I	The cubic boundary: sharp volume laws, uniform expansions, inverse charts and a Poisson law	5
1	The OEIS problem, the source audit, and the scope of the results	5
1.1	One fundamental family, rather than unrelated sequences	5
1.2	What is classical, and what this report actually adds	5
1.3	Relation to the ProveIt repository	6
1.4	Provenance and place in the collection	6
2	An elementary sharp criterion	7
3	A uniform centered expansion to every fixed order	8
3.1	Centering and the coefficient recurrence	8
3.2	The minor-arc estimate, including very large N	9
3.3	Proof of the uniform theorem	10
4	Critical constants and explicit OEIS expansions	11
4.1	A general all-order logarithmic expansion	11
4.2	Five corrections for A238608	12
4.3	Higher power columns and an important rounding issue	12
5	Dilute expansions and genuine exponential sectors	13
5.1	The uncentered coefficient system	13
5.2	Exponential and superexponential specializations	14
6	Inverting the asymptotic expansions	15
6.1	What inversion means for an integer sequence	15
6.2	The cubic family: Lambert W and two explicit corrections	15
6.3	Exponential columns: algebraic and invisible corrections	16
6.4	The diagonal sequence A238000	17
7	A Poisson boundary law in the conjugate partition model	18
7.1	Which repeated parts are being counted	18
7.2	A deletion estimate and the limit distribution	18
7.3	An explicit correction to the Poisson generating function	19
8	Exact computation and reproducible diagnostics	20
8.1	Independent coefficient extraction	20
8.2	Accuracy of the cubic expansion	21
8.3	Probability and inverse diagnostics	21
8.4	Reproduction and package contents	22
9	A formalization route from the existing Lean foundation	22
10	Further research questions and concrete next steps	23
11	Conclusion	24

II Beyond the cubic boundary: uniform asymptotics for $N \geq am^2$, a crossover hierarchy and analytic inversion	25
12 Provenance, scope and notation of Part II	25
12.1 Provenance	25
12.2 What Part II adds, and what it proves again	26
12.3 Notation	26
13 The research question and its precise scope	27
13.1 Why this problem was selected	27
13.2 Historical boundary and novelty discipline	29
13.3 Notation and meaning of an all-order statement	29
14 Main results	30
15 Exact generating functions and a finite-saddle expansion	32
16 A minor-arc estimate that survives an arbitrarily small saddle	33
16.1 Completion of the finite-saddle proof	34
17 Centering and the continuum saddle	35
18 Uniform Euler–Maclaurin analysis and a complete coefficient rule	35
18.1 A nonsingular local coordinate	36
18.2 An explicit all-order operator	36
18.3 Proof of the uniform theorem	37
19 Explicit coefficients and internal consistency checks	39
19.1 The entropy coefficients	39
19.2 The first finite-size functions	39
19.3 An independent check from the first central wave	40
20 The crossover hierarchy and its infinitude	40
20.1 A uniform dilute expansion	41
20.2 Necessity in the sharp centered boundary	41
20.3 Why the hierarchy does not terminate	41
21 How the quadratic and cubic OEIS formulas fit together	42
21.1 The classical quadratic leading formula	42
21.2 Recovery of the uncentered cubic boundary	42
21.3 What the intermediate scales add	43
22 Analytic inversion and discrete thresholds	43
22.1 The leading inverse map	43
22.2 First and all higher corrections	43
22.3 A safe integer-threshold statement	44
22.4 Lambert- W inversion along a power-law slice	45
23 What kind of transseries has been obtained?	46

24 Computational verification and reproducibility	46
24.1 Exact computations	46
24.2 Uniform-profile diagnostics	47
24.3 Crossover and inverse diagnostics	47
24.4 Package contents and commands	48
25 Further questions and research	49
25.1 Are all entropy coefficients strictly negative?	49
25.2 What controls the large-order behavior of $\mathbf{d}_r$?	49
25.3 Can the opposite moving-saddle endpoint be made uniform?	50
25.4 Can individual exponential waves be resolved?	50
25.5 Can the error constants support exact inversion?	50
25.6 How general is the centering hierarchy?	50
25.7 Can the analytic proof be formalized efficiently?	51
25.8 Convergence, Borel summability and optimal truncation of the $1/m$ series	51
25.9 When does the second-transition sign appear?	51
26 Conclusion of Part II	51
Appendix to Part I	52
A A compact all-order coefficient recipe	52

Part I

The cubic boundary: sharp volume laws, uniform expansions, inverse charts and a Poisson law

1 The OEIS problem, the source audit, and the scope of the results

1.1 One fundamental family, rather than unrelated sequences

Throughout, m and N are positive integers and $m \rightarrow \infty$. Define

$$F_m(q) = \prod_{j=1}^m \frac{1}{1 - q^j}, \quad p_m(N) = [q^N]F_m(q), \quad V_m(N) = \frac{N^{m-1}}{m!(m-1)!}. \quad (1.1)$$

The equivalence between the two partition interpretations follows by conjugating Ferrers diagrams. This is a standard, widely useful family of restricted partitions, not an isolated artificial recurrence. The same analytic problem occurs when the size N grows polynomially, exponentially, or factorially with the allowed number m of parts.

The following are the specific OEIS targets inspected for this report. The descriptions and posted leading formulas are recorded in the respective entries [1, 2, 3, 4].

Entry or family	Count	Role in this report
A238016	$p_m(m^k)$, an array	A posted general conjecture; cubic threshold
A238608	$p_m(m^3)$	Five corrections and inverse expansion
A258302–A258305	$p_m(cm^3)$, $c = 2, 3, 4, 5$	Uniform critical-scale family
A238609–A238615	$p_m(m^k)$, $4 \leq k \leq 10$	All-order power-column expansions
A238010; A237998	$p_m(q^m)$; $q = 2$	Exponential sectors and their inversion
A238000	$p_m(m^m)$	Superexponential sectors and a Lambert chart
A258670	$p_m((2m)!)$	Factorial-scale consequence

The general condition in A238016 is expressed using the nonstandard notation $f(m) \geq O(m^4)$. Interpreting the intended assertion as growth at least a positive constant times m^4 , it asserts $p_m(f(m)) \sim V_m(f(m))$. The exact criterion in Theorem 2.3 is instead $f(m)/m^3 \rightarrow \infty$. It covers, for example, $f(m) = \lceil m^3 \log m \rceil$, not merely quartic growth.

1.2 What is classical, and what this report actually adds

A condition remaining conjectural on an OEIS page does not establish that it is an open problem in the research literature. Canfield’s account of Szekeres’ formula records the Erdős–Lehner estimate

$$p_m(N) = \frac{1}{m!} \binom{N-1}{m-1} \exp\left(O(m^3/N)\right), \quad m = O(N^{1/3}), \quad (1.2)$$

in Step 5 and explains its composition-counting origin [5, pp. 4, 9–10]. In particular, the sufficiency of $N/m^3 \rightarrow \infty$ is already a classical consequence. The critical leading constant for $N = cm^3$ is explicitly posted in A238608 [2].

The polynomial part of a restricted partition function, called its first Sylvester wave, is also classical. Dilcher and Vignat give explicit formulas using higher-order Bernoulli polynomials [6]. Sills and Zeilberger give an algorithmic quasipolynomial approach [7]; O’Sullivan studies waves and their asymptotics in a different joint-growth regime [8].

The contribution here is a complete, reusable treatment of the following more precise package: finite elementary inequalities proving necessity as well as sufficiency; a uniform all-order theorem with all other arcs controlled; explicit critical coefficients through order five; an expansion with errors small enough to resolve q^{-m} sectors; explicit inverse charts; and a probabilistic boundary law with its first correction. The retrieved sources were not found to list these particular higher coefficients or this particular probability correction. That limited observation is not an exhaustive bibliographic novelty certification.

1.3 Relation to the ProveIt repository

The inspected repository file

```
Analysis/FabiusFunction/Lean/FabiusFunction/
PartitionBoundedParts.lean
```

provides a genuine formal starting point [12]. In namespace `Fabius`, it defines `boundedCount`, identifies restricted partitions with multiplicity vectors through `card_restricted_le_eq`, and proves the finite-product generating function in `hasSum_boundedCount_mul_pow`. The present analytic arguments can begin from precisely that generating function. We inspected the source but did not run the repository’s Lean build; none of the new asymptotic or probabilistic theorems is represented as already formalized.

1.4 Provenance and place in the collection

[write] **Added 1 October 2026, batch 74.** This report is a single manuscript, printed in full: manuscript 05 of batch 74 of ProveIt’s incoming-reports intake, delivered as the archive `OEIS_Restricted_Partitions_Cubic_Boundary.zip` in commit `c664fc2f0` and placed in the research-report collection, among the OEIS-sequence asymptotics reports, in commit `b669cff87`. It names no ProveIt commit: its repository input was a direct reading of the main-branch file `PartitionBoundedParts.lean`. At the placement commit that file still has the namespace `Fabius` (line 22) and the declarations `card_restricted_le_eq` (line 57), `boundedCount` (line 85) and `hasSum_boundedCount_mul_pow` (line 90), as described above. Those declarations are formal; nothing in this report is. Placement in the collection, beside that Lean development, confers no formal status on any statement here. The writing step changed only the labels (every label now carries the prefix `rpc:`), added these [write] notes, three labels and two references [10, 11]; no statement or proof was altered.

The nearest report in the collection is `a097356-sqrt-restricted-partitions` [11], on the same counts $p_m(N)$ at the quadratic scale $N \asymp m^2$ (its parameter is $\alpha = N/m^2$, held in a compact set). This report works at $N \asymp m^3$ and beyond, which is the end $\alpha \rightarrow \infty$ of that report’s research question “Uniform limits as the saddle moves to zero or infinity”. It does not answer that question: no expansion here is matched uniformly to the quadratic regime, and the two reports share no proposition. The partition reports `a033552-catalan-partitions` (parts restricted to Catalan numbers) and `a022629-distinct-partition-norms` (the products $\prod(1 + k^a q^k)$) concern different products. Sufficiency in the cubic criterion is the classical Erdős–Lehner estimate, as §1 says; the leading constant of A238608 is posted in that entry.

[write] **Added 5 October 2026, batch 98: Part II.** The note above describes Part I, the whole report as written in batch 74. Since batch 98 the report also has Part II, built from a second manuscript (§12.1). Its Theorem 14.1 is one expansion uniform for every $N \geq am^2$, so it is matched uniformly to the compact- α

regime of the neighbouring report; it answers the end $\alpha \rightarrow \infty$ of that report's question "Uniform limits as the saddle moves to zero or infinity", and proves that report's leading constant again (21.2). The sentence "It does not answer that question" remains true of Part I alone. The end $\alpha \rightarrow 0$ stays open (§25.3).

2 An elementary sharp criterion

Write $R_m(N) = p_m(N)/V_m(N)$. The scale m^3 can be detected without saddle points, Bernoulli numbers, or complex analysis.

Lemma 2.1 (Lattice-volume bounds). *For $m \geq 1$ and $N \geq 1$, put $A_m = m(m+1)/2 - 1$. Then*

$$1 \leq R_m(N) \leq \left(1 + \frac{A_m}{N}\right)^{m-1}. \quad (2.1)$$

Proof. The multiplicity of the part 1 is determined by the other multiplicities. Consequently $p_m(N)$ is the number of integer points in

$$\mathcal{S}_N = \left\{ (x_2, \dots, x_m) \in \mathbb{R}_{\geq 0}^{m-1} : \sum_{j=2}^m jx_j \leq N \right\}.$$

This simplex has volume $N^{m-1}/((m-1)! \prod_{j=2}^m j) = V_m(N)$. Every point of $\mathcal{S}_N$ lies in the unit box based at its coordinatewise floor, a feasible integer point. Conversely, every unit box based at a feasible point lies in $\mathcal{S}_{N+A_m}$. The boxes have disjoint interiors and unit volume. Taking volumes proves both bounds. For $m = 1$ the statement is the zero-dimensional identity $p_1(N) = 1$. $\square$

Lemma 2.2 (A transposition lower bound). *For $m \geq 3$ and $N \geq 1$,*

$$R_m(N) \geq \prod_{r=1}^{m-1} \left(1 + \frac{r}{N}\right) + \frac{m(m-1)^2}{4N} \geq 1 + \frac{m(m-1)^2}{4N}. \quad (2.2)$$

Proof. Let the symmetric group act on weak compositions $(x_1, \dots, x_m)$ of N by permuting coordinates. Its orbits are partitions into at most m parts. Burnside's identity gives

$$p_m(N) = \frac{1}{m!} \sum_{\sigma \in S_m} |\text{Fix}(\sigma)|.$$

The identity fixes $\binom{N+m-1}{m-1}$ compositions, which contributes the product in (2.2) after normalization by $V_m(N)$.

A transposition requires two coordinates to be equal. Its fixed-point count is the number of solutions of

$$2y + z_1 + \dots + z_{m-2} = N, \quad y, z_i \geq 0.$$

Eliminate z_{m-2} and apply the lower unit-box argument to the resulting simplex in dimension $m-2$. The count is at least $N^{m-2}/(2(m-2)!)$. There are $\binom{m}{2}$ transpositions. Their total contribution, divided by $m!V_m(N)$, is $m(m-1)^2/(4N)$. All other terms in Burnside's sum are nonnegative. $\square$

Theorem 2.3 (Exact threshold). *For every positive-integer sequence $N = N(m)$,*

$$p_m(N) \sim V_m(N) \iff N/m^3 \rightarrow \infty. \quad (2.3)$$

Moreover, when $m^3/N \rightarrow 0$, $R_m(N) = 1 + O(m^3/N)$, whereas $N/m^3 \rightarrow 0$ implies $R_m(N) \rightarrow \infty$.

Proof. By (2.1), $0 \leq \log R_m(N) \leq (m-1)A_m/N = O(m^3/N)$. This proves sufficiency and its error estimate. If $R_m(N) \rightarrow 1$, (2.2) forces $m(m-1)^2/N \rightarrow 0$, which is equivalent to $m^3/N \rightarrow 0$. The same lower bound proves the last assertion. $\square$

A useful extension of the lower bound also explains the critical exponential. For $0 \leq r \leq \lfloor (m-1)/2 \rfloor$, retain the permutations consisting of r disjoint transpositions and $m-2r$ fixed points. The identical simplex argument gives

$$R_m(N) \geq \sum_{r=0}^{\lfloor (m-1)/2 \rfloor} \frac{m^{2r} (m-1)^r}{4^r r! N^r}. \quad (2.4)$$

Here $x^r = x(x-1)\cdots(x-r+1)$ and $x^0 = 1$. For $N \sim cm^3$, any fixed finite truncation tends to the corresponding truncation of $\exp(1/(4c))$. Thus

$$\liminf_{m \rightarrow \infty} R_m(N) \geq e^{1/(4c)}.$$

An upper bound with the same constant needs more information; it is supplied by the uniform expansion below. Formula (2.4) is not itself an assertion that all remaining permutation types can simply be discarded.

3 A uniform centered expansion to every fixed order

3.1 Centering and the coefficient recurrence

Set

$$S_r(m) = \sum_{j=1}^m j^r, \quad M = N + \frac{S_1(m)}{2} = N + \frac{m(m+1)}{4}. \quad (3.1)$$

The exact factorization

$$e^{Nt} F_m(e^{-t}) = \frac{e^{Mt}}{m! t^m} \mathcal{A}_m(t), \quad \mathcal{A}_m(t) = \prod_{j=1}^m \frac{jt}{2 \sinh(jt/2)} \quad (3.2)$$

is the reason for choosing this center. The remaining factor is even. Its logarithm, in a neighborhood of zero, is

$$\log \mathcal{A}_m(t) = \sum_{r \geq 1} \gamma_r(m) t^{2r}, \quad \gamma_r(m) = -\frac{B_{2r} S_{2r}(m)}{2r(2r)!}. \quad (3.3)$$

We use $B_2 = 1/6$, $B_4 = -1/30$, and $B_6 = 1/42$. Formula (3.3) follows by differentiating $\log(z/(2 \sinh(z/2)))$ and integrating its Bernoulli expansion at zero.

Define $a_j(m)$ by $\mathcal{A}_m(t) = \sum_{j \geq 0} a_j(m) t^{2j}$. Then

$$a_0 = 1, \quad ja_j = \sum_{r=1}^j r \gamma_r a_{j-r}. \quad (3.4)$$

The first coefficients are

$$a_1 = -\frac{S_2}{24}, \quad (3.5)$$

$$a_2 = \frac{S_2^2}{1152} + \frac{S_4}{2880}, \quad (3.6)$$

$$a_3 = -\frac{S_2^3}{82944} - \frac{S_2 S_4}{69120} - \frac{S_6}{181440}. \quad (3.7)$$

Each a_j is a rational polynomial in m of degree at most $3j$. Indeed S_{2r} has degree $2r + 1 \leq 3r$, and the recurrence preserves the bound.

Theorem 3.1 (Uniform centered all-order expansion). *Fix $c_0 > 0$ and an integer $K \geq 0$. Uniformly over integer pairs with $N \geq c_0 m^3$, as $m \rightarrow \infty$,*

$$p_m(N) = \frac{M^{m-1}}{m!(m-1)!} \left\{ \sum_{j=0}^K \frac{a_j(m)(m-1)^{2j}}{M^{2j}} + O_{K,c_0}(m^{-K-1}) \right\}. \quad (3.8)$$

In particular, the j th displayed term is $O_{j,c_0}(m^{-j})$. The theorem concerns fixed truncation orders; it does not assert convergence of the resulting infinite asymptotic series.

3.2 The minor-arc estimate, including very large N

We give the details because a formal expansion at $q = 1$ alone does not justify (3.8). Other roots of unity must be controlled uniformly while the number of factors grows.

Lemma 3.2 (Decay on a Cauchy circle). *Let $0 < \tau \leq C_0 m^{-2}$, and put*

$$\rho_m(\theta) = \frac{|F_m(e^{-\tau-i\theta})|}{F_m(e^{-\tau})}.$$

For all sufficiently large m there are positive constants b, δ, C , with dependence on C_0 allowed, such that

$$\rho_m(\theta) \leq \left(1 + b(\theta/\tau)^2\right)^{-m/2}, \quad |\theta| \leq 1/m, \quad (3.9)$$

$$\rho_m(\theta) \leq (Cm\tau)^{\delta m}, \quad 1/m \leq |\theta| \leq \pi. \quad (3.10)$$

Proof. For each factor,

$$\frac{1 - e^{-j\tau}}{|1 - e^{-j\tau - ij\theta}|} = \left(1 + \frac{4e^{-j\tau} \sin^2(j\theta/2)}{(1 - e^{-j\tau})^2}\right)^{-1/2}.$$

If $|\theta| \leq 1/m$, then $|j\theta| \leq 1$ and $j\tau = O(1/m)$. The inequalities $|\sin(j\theta/2)| \geq cj|\theta|$ and $1 - e^{-j\tau} \leq j\tau$ prove (3.9) after multiplication.

For (3.10), a fixed positive fraction of the indices $j \leq m$ have $|\sin(j\theta/2)|$ bounded away from zero, uniformly in this interval. Here is an explicit justification. For $1/m \leq |\theta| \leq 4\pi/m$, choose j between $m/(16\pi)$ and $m/(8\pi)$; the arguments stay in a compact subinterval of $(0, \pi)$. For $4\pi/m \leq |\theta| \leq \pi$, the geometric-sum bound gives

$$\sum_{j=1}^m \sin^2(j\theta/2) \geq \frac{m}{2} - \frac{1}{2|\sin(\theta/2)|} \geq \frac{3m}{8}.$$

This forces a positive fraction of the summands to exceed a fixed positive constant. Each corresponding factor is at most $Cm\tau$; every other factor is at most one. Taking the minimum of the constants from the two cases proves the assertion. $\square$

[write] **Added 5 October 2026, batch 98.** Lemma 16.1 of Part II is (3.9) with the same proof under the weaker hypothesis $\tau \leq \bar{s}/m$; Lemma 16.2 bounds the integral over $1/m \leq |\theta| \leq \pi$ by $O(\tau(1 + \log m)e^{-cm})$ for every $\tau \leq \bar{s}/m$, by reserving the factor $j = 1$.

3.3 Proof of the uniform theorem

Proof of Theorem 3.1. Cauchy's formula, with $q = e^{-t}$, is

$$p_m(N) = \frac{1}{2\pi} \int_{-\pi}^{\pi} e^{N(\tau+i\theta)} F_m(e^{-\tau-i\theta}) d\theta. \quad (3.11)$$

Choose $\tau = m/M$ and the central interval $|\theta| \leq \tau m^{-2/5}$. We use three estimates.

Analytic Taylor remainder. For a sufficiently small absolute $\varepsilon > 0$, $\mathcal{A}_m(t)$ is analytic and uniformly bounded on $|t| \leq \varepsilon m^{-3/2}$. In fact $j|t| = O(m^{-1/2})$ there, and the sum of the absolute logarithmic bounds is $O(S_2(m)|t|^2) = O(1)$. Cauchy's coefficient estimate and evenness imply

$$\mathcal{A}_m(t) = \sum_{j=0}^K a_j(m) t^{2j} + O_K(m^{3K+3} |t|^{2K+2})$$

inside half that disk. The central interval is contained in this smaller disk for all sufficiently large m , uniformly in $N \geq c_0 m^3$.

Integrating the central remainder. For fixed nonnegative r and $m > r + 2$,

$$\int_{-\infty}^{\infty} |\tau + i\theta|^{-m+r} d\theta = \tau^{1-m+r} \sqrt{\pi} \frac{\Gamma((m-r-1)/2)}{\Gamma((m-r)/2)}.$$

The gamma quotient is $O_r(m^{-1/2})$. Together with Stirling's formula, this shows that integrating the Taylor remainder in (3.2), relative to $M^{m-1}/(m!(m-1)!)$, costs at most

$$O_K(m^{3K+3} \tau^{2K+2}) = O_{K,c_0}(m^{-K-1}).$$

The elementary inverse-Laplace identity

$$\frac{1}{2\pi i} \int_{\tau-i\infty}^{\tau+i\infty} e^{Mt} t^{-d} dt = \frac{M^{d-1}}{(d-1)!}, \quad d \geq 2,$$

follows by closing the contour to the left and taking the residue at zero. Apply it with $d = m - 2j$. Extending each central polynomial integral to this line changes it by an exponentially small quantity: the gamma kernel outside $|\theta| \leq \tau m^{-2/5}$ has a tail bounded by $\exp(-b_1 m^{1/5})$ times its natural integral size. This gives exactly the falling factorial in (3.8).

Discarding the rest of the original circle. On $\tau m^{-2/5} \leq |\theta| \leq 1/m$, (3.9) and the change of variable $v = \theta/\tau$ give a relative contribution $O(\exp(-b_2 m^{1/5}))$. On the remaining arcs, the relative contribution is bounded by

$$C \frac{\sqrt{m}}{\tau} (Cm\tau)^{\delta m}. \quad (3.12)$$

The factor $\sqrt{m}/\tau$ comes from normalizing the real-axis integrand by its gamma-kernel integral. It cannot be ignored when N is extremely large. For $\delta m > 1$, the right side is increasing in τ . Substituting $\tau \leq C_0/m^2$ bounds it by $m^{5/2}(C_1/m)^{\delta m}$, which is smaller than every fixed inverse power of m . Thus the argument is uniform even for factorial or faster growth of N , rather than merely for N bounded by a power of m .

Combining the three estimates proves (3.8). □

Remark 3.3 (What this theorem does not assert). For fixed m , the polynomial part of $p_m(N)$ is its first Sylvester wave. The coefficient algebra above is consistent with that classical polynomial. The theorem gives uniform finite-order asymptotics of the entire count, not just of a polynomial formally extracted at one singularity. It does not identify each exponentially small root-of-unity wave, provide a Stokes analysis, or specify an optimal truncation order growing with m .

[write] **Added 5 October 2026, batch 98.** Part II proves this theorem again, in logarithmic form, as the case $\varepsilon = m^2/M \leq 1/(c_0 m)$ of Theorem 14.1, which holds uniformly for every $N \geq am^2$ (see the note after that theorem). Its minor-arc Lemma 16.2 replaces (3.10) when $\tau \asymp 1/m$, where the bound $(Cm\tau)^{\delta m}$ above is useless.

4 Critical constants and explicit OEIS expansions

Corollary 4.1 (Critical crossover). *If $N/m^3 \rightarrow c \in (0, \infty)$, then*

$$\frac{p_m(N)}{V_m(N)} \longrightarrow e^{1/(4c)}. \quad (4.1)$$

Proof. The $K = 0$ case of Theorem 3.1 is $p_m(N) = M^{m-1}/(m!(m-1)!(1 + O(1/m)))$. Furthermore,

$$(m-1) \log(M/N) = (m-1) \log \left(1 + \frac{m(m+1)}{4N} \right) \longrightarrow \frac{1}{4c}.$$

□

[write] **Added 5 October 2026, batch 98.** A second route: Part II derives (4.1) from its uniform theorem (21.4), and the first correction $-61/(288m)$ of (4.11) below as (21.5), checked by its symbolic script. In Part II's centered normalization the cubic scale lies far above the first centered threshold $M \asymp m^{5/2}$ (Theorem 14.2).

4.1 A general all-order logarithmic expansion

Write $c = N/m^3$; this notation is permitted even when c varies with m . For every fixed K , uniformly for $c \geq c_0 > 0$,

$$\begin{aligned} \log p_m(cm^3) &= (m-3) \log m + 2m + (m-1) \log c - \log(2\pi) + \frac{1}{4c} \\ &\quad + \sum_{r=1}^K \frac{L_r(c)}{m^r} + O_{K,c_0}(m^{-K-1}). \end{aligned} \quad (4.2)$$

Here and below the left side is used only when cm^3 is an integer. The $L_r(c)$ are rational polynomials in $1/c$. Equivalently,

$$p_m(cm^3) = \frac{e^{2m+1/(4c)} c^{m-1} m^{m-3}}{2\pi} \left(\sum_{r=0}^K \frac{b_r(c)}{m^r} + O_{K,c_0}(m^{-K-1}) \right), \quad (4.3)$$

where

$$b_0 = 1, \quad kb_k = \sum_{r=1}^k r L_r b_{k-r}. \quad (4.4)$$

To obtain (4.2), take the logarithm of (3.8), substitute $M = cm^3(1 + (m^{-1} + m^{-2})/(4c))$, and use the classical Stirling expansion [9]:

$$\log \Gamma(m+1) + \log \Gamma(m) \sim 2m \log m - 2m + \log(2\pi) + 2 \sum_{r \geq 1} \frac{B_{2r}}{2r(2r-1)m^{2r-1}}. \quad (4.5)$$

Every operation is justified to a fixed order by Theorem 3.1. Thus the coefficient algorithm is backed by an asymptotic remainder theorem, not only by formal manipulation.

The first five logarithmic coefficients are

$$L_1(c) = -\frac{48c^2 + 13}{288c^2}, \quad (4.6)$$

$$L_2(c) = -\frac{144c^2 + 6c - 7}{576c^3}, \quad (4.7)$$

$$L_3(c) = \frac{11520c^4 + 122400c^2 + 14400c - 8081}{2073600c^4}, \quad (4.8)$$

$$L_4(c) = \frac{21600c^3 - 50400c^2 - 7855c + 2861}{2073600c^5}, \quad (4.9)$$

$$L_5(c) = -\frac{1}{21946982400c^6} (34836480c^6 + 304819200c^4 + 304819200c^3 - 235282320c^2 - 42928704c + 11425901). \quad (4.10)$$

As a short independent check of the first coefficient, the centered shift contributes $-1/(32c^2m)$, the a_1 term contributes $-1/(72c^2m)$, and Stirling contributes $-1/(6m)$. Their sum is (4.6).

4.2 Five corrections for A238608

At $c = 1$, the result becomes

$$p_m(m^3) = \frac{e^{2m+1/4}m^{m-3}}{2\pi} \left[1 - \frac{61}{288m} - \frac{37463}{165888m^2} + \frac{425075267}{3583180800m^3} - \frac{21833043287}{4127824281600m^4} - \frac{2048618007321713}{58251856261939200m^5} + O(m^{-6}) \right]. \quad (4.11)$$

The leading factor is the one already posted in A238608 [2]. The additional terms and their remainder are the refinement. The logarithmic form is often more stable numerically:

$$\log \left(\frac{2\pi p_m(m^3)}{e^{2m+1/4}m^{m-3}} \right) = -\frac{61}{288m} - \frac{143}{576m^2} + \frac{140239}{2073600m^3} - \frac{16897}{1036800m^4} - \frac{377689757}{21946982400m^5} + O(m^{-6}). \quad (4.12)$$

Substituting $c = 2, 3, 4, 5$ in the general formulas immediately gives the same order of refinement for A258302–A258305, without separate saddle analyses.

4.3 Higher power columns and an important rounding issue

For any fixed integer $k \geq 4$, substitute $c = m^{k-3}$ in (4.2). One obtains the complete inverse-power expansion of

$$p_m(m^k) \sim \frac{e^{2m}m^{(k-2)m-k}}{2\pi}, \quad (4.13)$$

the leading expression posted in the A238016 family [1]. For example, the quartic column A238609 satisfies

$$\begin{aligned} \log p_m(m^4) &= (2m - 4) \log m + 2m - \log(2\pi) + \frac{1}{12m} - \frac{139}{480m^3} \\ &\quad - \frac{1}{96m^4} + \frac{1403}{20160m^5} + O(m^{-6}). \end{aligned} \quad (4.14)$$

In particular, its first correction has the opposite sign from the cubic column's first correction.

If the intended count is $p_m(\lfloor cm^3 \rfloor)$ for a fixed nonintegral c , use $c_m = \lfloor cm^3 \rfloor / m^3$ in (4.2). Replacing c_m indiscriminately by c changes the logarithm by $O(m^{-2})$, which is large enough to corrupt coefficients beginning at the second correction. This is a discrete rounding effect, not a failure of the theorem.

5 Dilute expansions and genuine exponential sectors

The centered theorem is uniform in very large N , but its remainder $O(m^{-K-1})$ is not small enough to detect a term such as 2^{-m} . A separate expansion in $1/N$, with a remainder measured in m^3/N , is needed.

5.1 The uncentered coefficient system

Define

$$\mathcal{H}_m(t) = \prod_{j=1}^m \frac{jt}{1 - e^{-jt}} = \exp \left(\frac{S_1 t}{2} + \sum_{r \geq 1} \gamma_r t^{2r} \right) = \sum_{j \geq 0} h_j(m) t^j. \quad (5.1)$$

Put

$$\beta_j(m) = (m-1)^j h_j(m), \quad \beta_0 = 1. \quad (5.2)$$

If $\lambda_1 = S_1/2$, $\lambda_{2r} = \gamma_r$, and $\lambda_{2r+1} = 0$ for $r \geq 1$, then

$$jh_j = \sum_{r=1}^j r \lambda_r h_{j-r}. \quad (5.3)$$

Thus h_j and β_j have rational coefficients and degrees at most $2j$ and $3j$, respectively.

Theorem 5.1 (Uniform dilute expansion). *There is an absolute $\eta_0 > 0$ such that, for every fixed $K \geq 0$, uniformly for $0 < \eta = m^3/N \leq \eta_0$ and all sufficiently large m ,*

$$\frac{p_m(N)}{V_m(N)} = \sum_{j=0}^K \frac{\beta_j(m)}{N^j} + O_K((m^3/N)^{K+1}). \quad (5.4)$$

In particular, this holds along every sequence for which $m^3/N \rightarrow 0$.

Proof. Use (3.11) with $\tau = m/N = \eta/m^2$. The exact integrand now is $e^{Nt} \mathcal{H}_m(t) / (m! t^m)$. For a sufficiently small fixed $r > 0$, $\mathcal{H}_m$ is uniformly bounded on $|t| \leq r/m^2$: the linear logarithmic term is bounded, and all other terms have total absolute size $O(m^3|t|^2)$. Cauchy's estimate gives

$$\mathcal{H}_m(t) = \sum_{j=0}^K h_j t^j + O_K((m^2|t|)^{K+1})$$

on the disk of half this radius. Choose η_0 sufficiently small that τ lies well inside the disk and integrate over $|\theta| \leq r/(4m^2)$. The same gamma-kernel calculation as before bounds the relative remainder by $(m^2\tau)^{K+1} = \eta^{K+1}$. Each polynomial term gives $h_j(m-1)^j/N^j$ by inverse Laplace integration.

For completeness, the omitted arcs are small on this stronger scale, not merely small in $1/m$. On $r/(4m^2) \leq |\theta| \leq 1/m$, Lemma 3.2, after $v = \theta/\tau$, bounds the normalized tails by a polynomial factor in m times $(C\eta)^{m-1}$. For sufficiently small η_0 and m sufficiently large relative to K , this is $O_K(\eta^{K+1})$. On $1/m \leq |\theta| \leq \pi$, the analogous bound is

$$C \frac{m^{5/2}}{\eta} \left(\frac{C\eta}{m} \right)^{\delta m}.$$

After division by η^{K+1} , the positive remaining exponent of η allows substitution of η_0 , leaving a quantity tending rapidly to zero. These bounds also justify extending the polynomial integrals to the entire vertical line. This proves (5.4). $\square$

Taking the formal logarithm defines rational polynomials $D_j(m)$ through

$$D_j = \beta_j - \frac{1}{j} \sum_{r=1}^{j-1} r D_r \beta_{j-r}. \quad (5.5)$$

Theorem 5.1 justifies the corresponding logarithmic expansion with the same order of remainder. Its first terms are

$$D_1(m) = \frac{m(m^2 - 1)}{4}, \quad (5.6)$$

$$D_2(m) = -\frac{m(m^2 - 1)(13m^2 + 3m - 4)}{288}. \quad (5.7)$$

For example $h_2 = S_1^2/8 - S_2/24$, so $D_2 = (m-1)(m-2)h_2 - D_1^2/2$, an immediate exact check.

5.2 Exponential and superexponential specializations

For a fixed integer $q \geq 2$, put $a_q(m) = p_m(q^m)$ and $b = \log q$. The full fixed-order expansion is

$$\begin{aligned} \log a_q(m) &= bm(m-1) - \log \Gamma(m+1) - \log \Gamma(m) \\ &\quad + \sum_{j=1}^K D_j(m) e^{-jbm} + O_K\left(m^{3K+3} e^{-(K+1)bm}\right). \end{aligned} \quad (5.8)$$

This is an expansion in genuine exponentially separated sectors, rather than only a power series in $1/m$ renamed a transseries. The exact gamma background is retained so that its algebraic truncation error does not obscure the exponential corrections. In particular,

$$\begin{aligned} \log a_q(m) &= bm(m-1) - \log \Gamma(m+1) - \log \Gamma(m) \\ &\quad + \frac{m(m^2 - 1)}{4} q^{-m} - \frac{m(m^2 - 1)(13m^2 + 3m - 4)}{288} q^{-2m} + O(m^9 q^{-3m}). \end{aligned} \quad (5.9)$$

The leading term corresponds to A238010 and its columns, including A237998 when $q = 2$ [3].

For A238000, put $N = m^m$ in Theorem 5.1. The sectors are $m^{-jm} = \exp(-jm \log m)$, with the same polynomial coefficients $D_j(m)$. For A258670, put $N = (2m)!$; its sectors are $((2m)!)^{-j}$ [4]. In particular,

$$p_m((2m)!) = \frac{((2m)!)^{m-1}}{m!(m-1)!} \left(1 + \frac{m(m^2-1)}{4(2m)!} + O\left(\frac{m^6}{((2m)!)^2}\right) \right).$$

There is no need to solve a new saddle equation for each such sequence. The full collection follows from one uniform dilute theorem.

6 Inverting the asymptotic expansions

6.1 What inversion means for an integer sequence

A sequence has no canonical smooth interpolation. We therefore use an *asymptotic inverse chart*: a smooth expression $I(y)$ that, at the exact sample values $y = a(m)$, recovers m with a specified asymptotic error. The same expression can approximate an integer threshold inverse $Q(y) = \min\{m : a(m) \geq y\}$, but generally only up to an $O(1)$ rounding ambiguity. An exponentially accurate sample-point chart does not magically supply an exponentially accurate integer-valued function for all real y .

Each fixed truncation of a forward logarithmic profile is smooth for large real x and can be inverted there. Raising the truncation order produces consistent inverse coefficients. At sample values the forward remainder, divided by the derivative of the leading profile, controls the inverse error. This is the precise sense of the all-order inversions used below.

[write] **Added 1 October 2026, batch 74: an instance of repository results.** The distinction drawn here between a sample-point chart and the integer threshold $Q(y)$, with its $O(1)$ rounding ambiguity, is the content of the staircase theorem `p0:thm:staircase` of the transseries-and-inversion volume [10]: $Q = \lceil \nu \rceil$ for any increasing interpolation ν , and a chart with error at most η determines Q exactly when the separation condition of its part (2) holds. The inversions of this section are instances of that volume's inversion apparatus, and no novelty is claimed for the method.

6.2 The cubic family: Lambert W and two explicit corrections

Fix $c > 0$, and consider $a_c(m) = p_m(cm^3)$ on admissible integer arguments. Write

$$b_c = 2 + \log c, \quad D_c = \frac{1}{4c} - \log(2\pi c).$$

Equation (4.2) has the form

$$\log a_c(m) = m \log m + b_c m - 3 \log m + D_c + \frac{L_1(c)}{m} + \frac{L_2(c)}{m^2} + \cdots. \quad (6.1)$$

For large y , set $L = \log y$ and

$$w = W(ce^2 L), \quad x_0 = \frac{L}{w}. \quad (6.2)$$

Here W is the positive real solution of $we^w = z$. Then $\log x_0 + b_c = w$, and consequently $x_0 \log x_0 + b_c x_0 = L$ exactly. Define

$$d_0 = \frac{3 \log x_0 - D_c}{w + 1}, \quad d_1 = \frac{3d_0 - d_0^2/2 - L_1(c)}{w + 1}. \quad (6.3)$$

Theorem 6.1 (Cubic inverse chart). *At $y = a_c(m)$, with c fixed,*

$$m = x_0 + d_0 + \frac{d_1}{x_0} + O_c(x_0^{-2}). \quad (6.4)$$

There is an all-order expansion $x_0 + \sum_{r \geq 0} d_r(w; c)x_0^{-r}$, in the fixed-order asymptotic sense. Its coefficients are obtained recursively by Taylor substitution in (6.1); they are rational functions of w with constants depending on c .

Proof. Let $f_0(x) = x \log x + b_c x$. At x_0 , one has $f'_0 = w + 1$ and $f''_0 = 1/x_0$. Substitute $x = x_0 + d_0 + d_1/x_0$ in (6.1). The constant residual is $(w + 1)d_0 - 3 \log x_0 + D_c$, and the coefficient of x_0^{-1} is $(w + 1)d_1 + d_0^2/2 - 3d_0 + L_1(c)$. Both vanish by (6.3). The remaining residual is $O_c(x_0^{-2})$, whereas the derivative is asymptotic to $w + 1$. The mean-value theorem proves the stated, slightly weaker, $O_c(x_0^{-2})$ inverse error.

At any further fixed order, the new unknown coefficient appears multiplied by $w + 1$; everything else is already known. Division by $w + 1$ completes an induction. Since $\log x_0 = w - b_c$, the recursion stays within rational functions of w . The forward expansion can be taken to arbitrarily high fixed order by Theorem 3.1. $\square$

For A238608 specifically,

$$d_0 = 3 - \frac{37/4 - \log(2\pi)}{w + 1}, \quad d_1 = \frac{3d_0 - d_0^2/2 + 61/288}{w + 1}, \quad w = W(e^2 \log y).$$

These corrections are important even for moderately large indices: the raw Lambert term can miss the index by nearly two while the corrected expression is already accurate to about 10^{-4} at $m = 60$.

[write] **Added 1 October 2026, batch 74: an instance of repository results.** The core $x \log x + b_c x = L$, solved by the principal branch as in (6.2), is the Lambert core of the inverse gamma function in `p6:thm:gamma` of the transseries-and-inversion volume [10] (there $T \log T - T = R$ with $T = R/W_0(R/e)$), shifted by the constant b_c ; the corrections $d_0, d_1, \dots$ are its recursive perturbed inversion `p0:thm:perturbed-inversion`. The exponential columns below use the same perturbed-inversion step with the small parameter e^{-bX} . Theorem 6.1 and (6.8) are instances of that apparatus.

6.3 Exponential columns: algebraic and invisible corrections

For $q \geq 2$ fixed, define the exact smooth background

$$G_q(x) = bx(x - 1) - \log \Gamma(x + 1) - \log \Gamma(x), \quad b = \log q. \quad (6.5)$$

Stirling's expansion yields

$$G_q(x) = bx^2 - 2x \log x + (2 - b)x - \log(2\pi) - \frac{1}{6x} + \frac{1}{180x^3} - \dots$$

Put

$$s = \sqrt{L/b}, \quad d = \frac{\log s}{b} + \frac{b - 2}{2b}, \quad e = \frac{d^2}{2} + \frac{d}{b} + \frac{\log(2\pi)}{2b}. \quad (6.6)$$

Taylor substitution gives the algebraic inverse chart

$$m = s + d + \frac{e}{s} + O_q\left(\frac{(\log s)^3}{s^2}\right), \quad y = a_q(m). \quad (6.7)$$

For example, the coefficient of s cancels because $2bd = 2\log s + b - 2$, and the constant residual cancels because $2be = bd^2 + 2d + \log(2\pi)$.

To expose an exponentially small inverse correction, let X be the exact large solution of $G_q(X) = \log y$. This background inverse is well defined for large y since

$$G'_q(X) = b(2X - 1) - \psi(X + 1) - \psi(X) \sim 2bX > 0.$$

Equation (5.8) and one implicit Taylor step give

$$m = X - \frac{D_1(X)e^{-bX}}{G'_q(X)} + O_q(X^5e^{-2bX}), \quad y = a_q(m). \quad (6.8)$$

The sign is negative: the positive count correction raises $\log a_q$ above the gamma background, so the corrected index at fixed y is smaller. The $O(m^6e^{-2bm})$ forward error divided by $G'_q(m) \asymp m$ gives the displayed safe inverse error; the Taylor cross terms are smaller.

All further exponential sectors can be computed recursively. For example, write $m = X + u_1(X)e^{-bX} + u_2(X)e^{-2bX} + \dots$. Then

$$u_1 = -D_1/G'_q, \quad (6.9)$$

$$u_2 = -\frac{D_2 + (D'_1 - bD_1)u_1 + G''_qu_1^2/2}{G'_q}. \quad (6.10)$$

Theorem 5.1 supplies the required forward remainder at every fixed sector order. Retaining only finitely many terms of Stirling in G_q would not justify (6.8): that algebraic error would swamp the quantity being resolved.

6.4 The diagonal sequence A238000

The same procedure gives a particularly simple leading inverse for A238000. Here

$$\log p_m(m^m) = (m^2 - 3m)\log m + 2m - \log(2\pi) + O(1/m),$$

with exponentially smaller corrections in m^{-m} supplied by Section 5. If

$$x_0 = \sqrt{\frac{2L}{W(2L)}},$$

then $x_0^2 \log x_0 = L$, and a first correction is

$$m = x_0 + \frac{3\log x_0 - 2}{2\log x_0 + 1} + O(x_0^{-1}) \quad \text{at } y = p_m(m^m). \quad (6.11)$$

This follows by dividing the leading residual $-3x_0 \log x_0 + 2x_0$ by the derivative $x_0(2\log x_0 + 1)$. It provides another inverse chart from the same partition theorem, rather than an unrelated inversion example.

7 A Poisson boundary law in the conjugate partition model

7.1 Which repeated parts are being counted

Choose a partition of N into *at most* m parts uniformly, and pad it with zeros to obtain

$$0 \leq x_1 \leq \cdots \leq x_m, \quad x_1 + \cdots + x_m = N.$$

This is the conjugate of the original bounded-part-size model. The distinction matters: partitions whose part sizes are bounded by m typically have many repeated small parts, and the theorem below is not a claim about those multiplicities.

Set $d_1 = x_1$ and $d_i = x_i - x_{i-1}$ for $i \geq 2$. Then

$$\sum_{i=1}^m (m - i + 1) d_i = N.$$

Let

$$Z_m = \#\{i : 2 \leq i \leq m, d_i = 0\},$$

the number of equal adjacent pairs in the padded partition. A gap equal to zero corresponds to a zero multiplicity for one of the weights $1, \dots, m-1$ in the generating product. Its probability generating function is exactly

$$\mathbb{E}[u^{Z_m}] = \frac{[q^N] F_m(q) \prod_{j=1}^{m-1} (u + (1-u)q^j)}{p_m(N)}. \quad (7.1)$$

Indeed for each marked weight the sum over a gap is $u + q^j + q^{2j} + \cdots = (u + (1-u)q^j)/(1-q^j)$.

7.2 A deletion estimate and the limit distribution

Lemma 7.1 (Deleting a fixed number of factors). *Fix k . Uniformly over k -element sets $S \subseteq \{1, \dots, m\}$ and N/m^3 in a compact subinterval of $(0, \infty)$,*

$$\frac{[q^N] F_m(q) \prod_{j \in S} (1 - q^j)}{p_m(N)} = \left(\prod_{j \in S} j \right) \frac{(m-1)^k}{M^k} (1 + O_k(m^{-1})). \quad (7.2)$$

Proof. On the central interval of Section 3,

$$\prod_{j \in S} (1 - e^{-jt}) = \left(\prod_{j \in S} j \right) t^k (1 + O_k(m|t|)) = \left(\prod_{j \in S} j \right) t^k (1 + O_k(m^{-1})).$$

Inserting this factor in the contour proof changes t^{-m} to t^{-m+k} , so the inverse-Laplace integral contributes $(m-1)^k/M^k$. Both $\mathcal{A}_m(t)$ and the denominator count have relative error $O(1/m)$ at this precision. On the other arcs the new product has modulus at most 2^k . Since k is fixed and $\tau \asymp m^{-2}$, normalizing by the smallest possible main deleted-factor contribution loses only a fixed power of m , harmless compared with the arc decay. All bounds are uniform in the choice of S . $\square$

Theorem 7.2 (Poisson repeated-part boundary). *If $N/m^3 \rightarrow c \in (0, \infty)$, then*

$$Z_m \xrightarrow{d} \text{Poisson}\left(\frac{1}{2c}\right). \quad (7.3)$$

With probability tending to one, the partition has exactly m positive parts and no part occurs three or more times. Thus Z_m also agrees, with probability tending to one, with the number of distinct positive part sizes that occur twice.

Proof. For every fixed k , the factorial-moment identity from (7.1) and Lemma 7.1 give

$$\mathbb{E}[Z_m^k] = k! \frac{(m-1)^k}{M^k} \sum_{\substack{S \subseteq \{1, \dots, m-1\} \\ |S|=k}} \prod_{j \in S} j \left(1 + O_k(m^{-1})\right).$$

For fixed k , the elementary symmetric sum here equals

$$\frac{1}{k!} \left(\sum_{j=1}^{m-1} j \right)^k (1 + O_k(m^{-1})).$$

To see this, expand the power and remove tuples with a repeated index; their relative total is bounded by a constant times $\sum j^2 / (\sum j)^2 = O(1/m)$. Therefore

$$\mathbb{E}[Z_m^k] \rightarrow (1/(2c))^k.$$

These are the factorial moments of the asserted Poisson law. For clarity, the first moment gives tightness; the next higher moments give uniform integrability of every fixed power along subsequences. Any subsequential limit therefore has the Poisson moments. The Poisson distribution is moment-determinate, as its moment generating function is finite in a neighborhood of zero. This proves (7.3).

The event $d_1 = 0$ is a single deletion of weight m , and hence has probability $m(m-1)/M(1 + O(1/m)) = O(1/m)$. A triple repetition requires two consecutive gaps to vanish. By (7.2), summing these two-deletion probabilities over at most m adjacent pairs costs

$$O\left(\frac{m^2}{M^2} \sum_{j=1}^{m-1} j(j+1)\right) = O(m^{-1}).$$

Outside these two exceptional events there are m positive parts and only double repetitions, proving the last assertion. $\square$

7.3 An explicit correction to the Poisson generating function

The limit can be refined without assuming independence of the gaps.

Theorem 7.3 (First finite-size probability correction). *Let $c = N/m^3$ range over a compact subinterval of $(0, \infty)$. Uniformly for real $0 \leq u \leq 1$,*

$$\mathbb{E}[u^{Z_m}] = e^{(u-1)/(2c)} \left[1 - \frac{1}{m} \left(\frac{u-1}{c} + \frac{7u(u-1)}{24c^2} \right) + O_c(m^{-2}) \right]. \quad (7.4)$$

The constant in the error can be chosen uniformly on the indicated compact c interval. In particular,

$$\mathbb{P}(Z_m = 0) = e^{-1/(2c)} \left(1 + \frac{1}{cm} + O_c(m^{-2}) \right). \quad (7.5)$$

Proof. Put $v = u - 1$, $S'_r = \sum_{j=1}^{m-1} j^r$, and $M_v = M + vS'_1$. Near zero,

$$\log \prod_{j=1}^{m-1} (u + (1-u)e^{-jt}) = vS'_1 t - \frac{v+v^2}{2} S'_2 t^2 + O(m^4 |t|^3),$$

uniformly for $0 \leq u \leq 1$. After combining this with (3.3), define

$$C_v = \frac{S_2}{24} + \frac{v+v^2}{2} S'_2.$$

The same centered contour proof, now at $\tau = m/M_v$, yields for the numerator of (7.1)

$$\frac{M_v^{m-1}}{m!(m-1)!} \left(1 - C_v \frac{(m-1)(m-2)}{M_v^2} + O(m^{-2}) \right).$$

The error follows from $m^4 |t|^3 + m^6 |t|^4 = O(m^{-2})$ on the central scale. The old arc estimates remain valid: for $0 \leq u \leq 1$ and $|q| \leq 1$, every multiplier $u + (1-u)q^j$ has modulus at most one. Also $M_v = M + O(m^2) > 0$ and the change of real-axis normalization costs only a bounded factor.

Divide by the same formula at $v = 0$ and take a logarithm. With $N = cm^3$,

$$\begin{aligned} (m-1) \log(M_v/M) &= \frac{v}{2c} - \frac{1}{m} \left(\frac{v}{c} + \frac{v+v^2}{8c^2} \right) + O(m^{-2}), \\ -C_v \frac{(m-1)(m-2)}{M_v^2} + C_0 \frac{(m-1)(m-2)}{M^2} &= -\frac{v+v^2}{6c^2 m} + O(m^{-2}). \end{aligned}$$

Since $1/8 + 1/6 = 7/24$, exponentiation proves (7.4). Setting $u = 0$ proves (7.5). $\square$

There is a useful independent exact identity at $u = 0$. Subtract the staircase $(0, 1, \dots, m-1)$ from a strictly increasing nonnegative partition. The result is an arbitrary weakly increasing nonnegative partition of $N - T$, where $T = m(m-1)/2$. Thus

$$\mathbb{P}(Z_m = 0) = \frac{p_m(N - T)}{p_m(N)}. \quad (7.6)$$

Inserting the centered expansion into this ratio checks both the limit $e^{-1/(2c)}$ and the first correction $1/(cm)$. Higher pgf corrections can be generated by retaining further terms in $\log(u + (1-u)e^{-z})$ and using the same coefficient recursions. We do not claim a total-variation rate or a uniform complex- u expansion from the real interval estimate alone.

8 Exact computation and reproducible diagnostics

8.1 Independent coefficient extraction

The package's `verify.py` computes integer coefficients by the standard product update

$$P_0(0) = 1, \quad P_0(n) = 0 \ (n > 0), \quad P_j(n) = P_{j-1}(n) + P_j(n-j),$$

where negative arguments contribute zero. A second implementation uses the logarithmic derivative of (1.1):

$$np_m(n) = \sum_{r=1}^n \left(\sum_{\substack{d|r \\ d \leq m}} d \right) p_m(n-r). \quad (8.1)$$

These two methods agree for $1 \leq m \leq 12$ and $0 \leq n \leq 200$. The finite inequalities in Section 2 are tested after clearing denominators, so their verification uses exact integer arithmetic. All displayed initial A238608 values through $m = 16$ agree with the inspected OEIS entry [2].

The complete run used 90-decimal-digit floating-point arithmetic only after the exact counts were produced. It performed 9,230 exact assertions and recorded 258 count samples, with $m \leq 60$ and coefficient arrays extending to $N = 320000$. The sample families include cm^3 for $1 \leq c \leq 5$, m^4 , and 2^m . These are finite tests; the all-order and uniform statements rest on the contour estimates, not on extrapolating the tables.

8.2 Accuracy of the cubic expansion

Let $A_K(m)$ be the right side of (4.11), truncated after m^{-K} . The following entries are signed relative errors $A_K(m)/p_m(m^3) - 1$ from the included exact computation.

m	$K = 0$	$K = 1$	$K = 3$	$K = 5$
10	$2.38780 \cdot 10^{-2}$	$2.19166 \cdot 10^{-3}$	$8.72320 \cdot 10^{-7}$	$-2.93140 \cdot 10^{-8}$
20	$1.12656 \cdot 10^{-2}$	$5.55991 \cdot 10^{-4}$	$4.40927 \cdot 10^{-8}$	$-4.51323 \cdot 10^{-10}$
40	$5.46413 \cdot 10^{-3}$	$1.40056 \cdot 10^{-4}$	$2.41567 \cdot 10^{-9}$	$-7.04814 \cdot 10^{-12}$
60	$3.60523 \cdot 10^{-3}$	$6.24068 \cdot 10^{-5}$	$4.54363 \cdot 10^{-10}$	$-6.18667 \cdot 10^{-13}$

The full CSV includes $K = 2$, the normalized exact count, and the other values of c . A small error at one index alone would not prove an expansion; these data are useful as stringent checks on coefficient signs, normalization, and omitted shifts.

8.3 Probability and inverse diagnostics

For $c = 1$, the limiting mean is $1/2$, the limiting zero probability is $e^{-1/2} \approx 0.60653066$, and the limiting pgf at $u = 1/2$ is $e^{-1/4} \approx 0.77880078$. Exact finite values are

m	$\mathbb{E}Z_m$	$\mathbb{P}(Z_m = 0)$	$\mathbb{E}[2^{-Z_m}]$
10	0.3846890512	0.6682085339	0.8213708910
20	0.4390680097	0.6371577383	0.8006057040
40	0.4686467899	0.6217743594	0.7898304769
60	0.4788932680	0.6166760199	0.7861818675

The first-corrected pgf at $u = 1/2$ has relative error about $7.04 \cdot 10^{-5}$ at $m = 60$, consistent with its $O(m^{-2})$ remainder. The exact pgf there is calculated using the coefficients of $\prod_{j=1}^{m-1} (1 + q^j)$, divided by 2^{m-1} , in (7.1).

At $y = p_{60}(60^3)$, the uncorrected Lambert chart has index error -1.96028 , while (6.4) has error $1.66805 \cdot 10^{-4}$. At $y = p_{18}(2^{18})$, the exact gamma-background inverse X has index error $2.99966 \cdot 10^{-4}$;

adding the exponential correction in (6.8) reduces it to $-1.72742 \cdot 10^{-8}$. These diagnostics illustrate why algebraic and exponential inverse corrections should be treated separately.

8.4 Reproduction and package contents

The archive contains the article source and PDF, plus:

derive_coefficients.py

Exact Bernoulli/power-sum derivation of $L_r(c)$ and $b_r(c)$, with a selectable fixed order. The default is five.

asymptotics.py

Forward logarithmic profiles, the explicit cubic inverse, numerical inversion of a chosen truncated profile, and the first exponential inverse correction.

verify.py

Exact product and divisor-recurrence checks, finite bounds, OEIS initial-value checks, probability diagnostics, and inverse diagnostics.

Data files

`coefficients.json`, `exact_values.json`,
`critical_errors.csv`, `poisson_checks.csv`,
`inverse_checks.csv`, `verification_summary.json`.

Only SymPy and mpmath are required beyond the Python standard library. The article uses standard L^AT_EX packages and builds with three runs of `pdflatex`. The README supplies the exact commands. No network access is required to reproduce the included calculations after dependencies are installed.

[write] **Added 1 October 2026, batch 74: the shipped layout.** In the collection the four programs (with `build.sh`) are in `code/` and the data files, `requirements.txt` and the two run records in `data/`; this source is `article.tex`, renamed from `restricted_partitions_cubic_boundary.tex`, and the delivered PDF is not shipped. Every program reads and writes next to itself, so `verify.py` cannot find `coefficients.json` from `code/`, and `build.sh` still names the old source file. Rerun on a flat copy of `code/` and `data/`; the README gives the commands.

9 A formalization route from the existing Lean foundation

The exact combinatorial foundation already present in ProveIt suggests a clean separation into finite and analytic layers. This is a proposed route, not a claim of completed code.

The first layer would formalize the weak-composition orbit model and the transposition fixed-point injection. These provide an entirely discrete proof of the lower obstruction $m(m-1)^2/(4N)$. A companion lattice-simplex volume lemma gives the upper bound and the exact threshold. The existing multiplicity-vector equivalence can supply the bridge between the two models.

The second layer would formalize coefficient extraction for the finite q -Pochhammer product, the analytic identity (3.2), and the Bernoulli recurrence (3.4). At each fixed order, the explicit coefficient identities reduce to rational polynomial algebra and are good candidates for small proof certificates.

The genuinely analytic work is then localized: a uniform trigonometric index-count lemma, a Cauchy-circle decay bound, gamma-kernel integral estimates, and a uniform analytic Taylor remainder on a shrinking disk. The distinction between a fixed parameter and a constant uniform over $N \geq c_0 m^3$ must be represented explicitly in the quantifiers. Finally, probability is built from finite uniform measures on partition sets, coefficient-deletion identities, and factorial moments.

The source file inspected in the repository is compatible with this program, but neither its availability nor the successful Python checks certify any of these future Lean developments. In particular, a formalization must not replace the minor-arc estimate with the assertion that $q = 1$ is the only relevant singularity: the other roots of unity are present and must be bounded.

10 Further research questions and concrete next steps

The results above leave substantial, sharply formulated problems. The following are proposals for further work, not assertions that every one is historically open in exactly this form.

1. Optimal truncation and individual waves. Determine how the best order in (3.8) grows with m , and compute the leading nontrivial root-of-unity contribution in the cubic regime. The first target is the $q = -1$ wave, followed by a uniform comparison of all remaining orders. This would turn the present fixed-order expansions into a substantially more complete transseries.

[write] **Added 5 October 2026, batch 98.** Part II poses the same question for $N \geq am^2$ (§25.4) and the optimal truncation of its expansion (§25.8).

2. A uniform bridge below the cubic scale. Extend the centered analysis into $m^{5/2} \lesssim N \ll m^3$, where the centering factor can be large and the quadratic Bernoulli correction must itself be exponentiated. Match the resulting expansion to a small-parameter expansion of the Szekeres saddle. The precise validity range and matching error are more useful targets than simply recovering a leading logarithm.

[write] **Added 5 October 2026, batch 98: answered by Part II.** Theorem 14.1 gives one fixed-order expansion of $\log(p_m(N)/V_m(M))$ with remainder $O_{a,J}(m^{-J-1})$ uniformly for every $N \geq am^2$, so it covers $m^{5/2} \lesssim N \ll m^3$ and the Szekeres regime $N \asymp m^2$ at once, with the matching built in (§20, §21). The exponentiated quadratic Bernoulli correction is the term $-m^5/(72M^2)$, and $M \asymp m^{5/2}$ is the exact threshold of the centered volume law (Theorem 14.2), followed by the scales $m^{2+1/(2r)}$ (Theorem 14.3).

3. Structural degree cancellation in logarithmic coefficients. The general construction only immediately gives $\deg D_j \leq 3j$, while D_1 and D_2 have degrees 3 and 5. Investigate a systematic sharper degree theorem, its leading coefficients, and a connected-object or cumulant interpretation. Such a result could reduce symbolic complexity and improve error estimates across many dilute sequences.

4. Higher probability corrections and quantitative convergence. Compute an all-order version of (7.4), then prove a uniform complex- u domain sufficient for extracting point probabilities with explicit error bounds. Obtain a total-variation estimate for the Poisson approximation, rather than inferring one from convergence of fixed moments.

5. Rare triple repetitions. The present proof bounds their probability by $O(1/m)$ at the cubic scale. Determine its leading constant and joint asymptotics with the number of doubletons. At other scales, identify when tripletons develop a nondegenerate limit and whether a hierarchy of multiplicity transitions results.

6. Deletions whose number grows. Lemma 7.1 assumes fixed k . Find the maximal growing range $k = k(m)$ for which a useful uniform coefficient ratio holds. This would give moderate-deviation information for Z_m , not just a Poisson limit at fixed counts.

7. Other weight systems. Replace $1, \dots, m$ by structured integer weights with an available unit weight and controlled power sums. The local expansion generalizes immediately, but the global arc estimate depends on the weights' arithmetic distribution. Find hypotheses that separate a universal volume/cumulant law from genuine periodic obstructions.

[write] **Added 5 October 2026, batch 98.** The same question for Part II's centering hierarchy is §25.6.

8. Certified inverse thresholds. Combine explicit finite constants in the remainder bounds with monotonicity to produce intervals enclosing the integer inverse $Q(y)$. A useful algorithm must report when its interval straddles two integers, rather than silently rounding a smooth chart. This is particularly important when y lies very close to an exact sequence value.

[write] **Added 5 October 2026, batch 98.** Part II extends the question to $N \geq am^2$ (§25.5, with Proposition 22.2).

9. A proof-producing asymptotic pipeline. Starting from the finite-product generating function, automate power sums, Bernoulli coefficients, fixed-order remainder parameters, transseries substitution, and inverse reversion. The symbolic output should be a small certificate checkable independently, with numerical fitting excluded from the proof path.

[write] **Added 5 October 2026, batch 98.** Part II's formalization question is §25.7.

10. OEIS and literature reconciliation. Prepare a narrowly scoped, source-backed update for the affected entries: replace the vague quartic-growth condition by a precise sufficient condition, attach the classical attribution, and separately record explicit refinements. A useful contribution is improved mathematical traceability, not relabeling classical consequences as breakthroughs.

11 Conclusion

The restricted-partition family admits one coherent explanation of several OEIS asymptotic patterns. The volume approximation has an exact cubic threshold, critical growth leaves a nontrivial constant, and larger sizes admit polynomial, exponential, or factorial-scale refinements governed by the same Bernoulli coefficient system. Uniform contour control is what turns formal local algebra into theorems about the whole counting sequence.

The inverse formulas illustrate a second important distinction: an exact gamma background is needed before exponentially small corrections can be meaningfully inverted. Finally, the conjugate partition model shows that the cubic boundary is not merely an artifact of asymptotic notation. It is

a regime where repeated parts survive in a nondegenerate Poisson law, with a computable finite-size correction. The elementary criterion, all-order expansions, inverse charts, and probabilistic law form the substantive proof package supplied by this report.

Part II

Beyond the cubic boundary: uniform asymptotics for $N \geq am^2$, a crossover hierarchy and analytic inversion

12 Provenance, scope and notation of Part II

[Added 5 October 2026, batch 98.] Part II was added in the write of batch 98. It prints manuscript 09 of that batch in full, as §§13–26, in Part I’s setting: the same counts $p_m(N)$, the same centering $X = N + m(m+1)/4$ (Part I’s M) and the same centered volume. It answers Part I’s research question 2 and the end $\alpha \rightarrow \infty$ of a research question of the neighbouring report `a097356-sqrt-restricted-partitions`, and it proves several of Part I’s results again; those are named, not reprinted (Table 2). Its symbols that clash with Part I’s are renamed (Table 3). Text added in the write is marked [write]; the manuscript’s statements, proofs, tables and numbers are otherwise unchanged.

12.1 Provenance

Manuscript 09 of batch 98 of ProveIt’s incoming-reports intake is the archive `OEIS_Restricted_Partitions_Crossover.zip` (454,491 bytes, 21 files under one directory `partition_crossover/`), which arrived in commit `2172df76a` (5 October 2026) and was placed in this report by commit `0fba5167f`. Its article is titled *Beyond the Cubic Boundary: Uniform restricted-partition asymptotics, an infinite crossover hierarchy, and analytic inversion* (20 pages, an “OEIS research report” dated 4 October 2026); its title page reads “Prepared for Vladimir Reshetnikov”, and its PDF metadata names “OpenAI; research report prepared for Vladimir Reshetnikov” as author. It describes itself as AI-generated, unrefereed and not formalized.

It names no ProveIt commit. Its source audit (`02-crossover-SOURCE_AUDIT.md`) read, through a GitHub connector on the default branch, the READMEs of this report and of `a097356-sqrt-restricted-partitions`, the collection README and selected search results. It reports the Git blob `ffeb1d4cf2ab...` for this report’s README: that is the README written in batch 74 (commit `f440ed3c0`), unchanged until this write, and the neighbouring report was unchanged since commit `31cf70f09`. So manuscript 09 saw both reports as they stood, but only their READMEs: it does not cite Part I’s theorems, lemmas or questions by number. The write added those pointers as dated notes. Its bibliography of eight entries is merged into this report’s: its A238016 and A238608 entries are [1, 2] (it inspected them on 4 October 2026), its Canfield, Sills–Zeilberger and O’Sullivan entries are [5, 7, 8] (it cites version 2, 2018, of O’Sullivan’s preprint, and records that the journal web page dates Canfield’s paper 21 November 1996, in the volume dated 1997), its entry for the neighbouring report is [11], and its entry for this report, which it says it used only for the scope

statement, became references to Part I. Its A097356 entry is [13].

Scope and non-claims of manuscript 09, kept. It is an AI-generated, unrefereed research manuscript with conventional proofs, exact finite checks and high-precision diagnostics; no Lean or Rocq verification was run, and its proofs “remain subject to expert review”. Szekeres’ leading asymptotics (Canfield), the uncentered cubic law, the leading A238608 law and its correction $-61/288$ are classical or already in the repository and are not claimed; the saddle method, Euler–Maclaurin summation, Lagrange inversion and Lambert- W inversion are classical tools. Its source audit was bounded, not exhaustive: “There may be older equivalent results or coefficients”, and no priority claim is made. Its expansions hold at each fixed order, with no uniformity in a growing order and no convergence, Gevrey bound, Borel summability or optimal truncation of the $1/m$ series; calling the result a complete resurgent transseries “would be too strong”, since the root-of-unity contributions, their actions, amplitudes and Stokes behaviour are not derived. It is not uniform as $N/m^2 \rightarrow 0$, and its endpoint statement at $\varepsilon \rightarrow 4$ concerns the entropy function only, not a partition regime. Nonvanishing and negativity of every d_r are not claimed. Its inverse charts are smooth asymptotic charts, not integer-rounding rules, with no explicit error constants; its numerics are 55-digit `mpmath` values, not interval certificates, and do not instantiate the remainder constants. Its second-transition samples ($m \leq 96$) have not reached the limiting sign. Its seven research questions are proposals. Its slice inversion is “an application of the repository’s transseries-inversion program, not a new inversion method”. It does not reprove or claim the floor-phase, three-exception or plateau results of the neighbouring report. It submitted nothing to the OEIS and committed nothing to the repository.

12.2 What Part II adds, and what it proves again

Table 1 compares the two Parts topic by topic. Table 2 lists the earlier results, of Part I and of the neighbouring report, that manuscript 09 proves again, and the route it takes; none of them is reprinted.

12.3 Notation

Part II keeps manuscript 09’s notation except where a symbol clashes with Part I or is used twice in the manuscript; Table 3 fixes the convention, and the renamed objects are written in calligraphic or sans-serif type. No normalization is changed. The centered size X is Part I’s M ; the letter X is kept because Part II’s statements are written in it. Unsubscripted c , C , c' , δ and their subscripted forms in proofs are generic constants, except in §21, where $N \sim cm^3$ uses Part I’s $c = N/m^3$.

Table 3: Notation of manuscript 09 against Part I. Part II uses the second column; the last column names the Part I symbol it must not be read as.

Notion	Part II	Manuscript 09	Not to be read as (Part I)
centered size $N + m(m+1)/4$	X	X	the same object as Part I’s M ; not the gamma-background root X of §6
centered volume	$V_m(X)$	$V_m(X)$	Part I’s $V_m(N)$ is the same function at N

Notion	Part II	Manuscript 09	Not to be read as (Part I)
m^2/X	ε	ε	the small constant ε in the proof of Theorem 3.1
fixed constant in $N \geq am^2$	a	a	$a_j(m)$, $a_q(m)$, $a_c(m)$
regularized logarithm	$h(z)$	$h(z)$	the coefficients $h_j(m)$ of $\mathcal{H}_m$
its mean $\int_0^1 h(sx) dx$	$H(s)$	$H(s)$	$\mathcal{H}_m(t)$
$1 - sH'(s)$	$B(s)$	$B(s)$	the Bernoulli numbers B_{2k} (used in both Parts)
$1 + s^2H''(s)$, s^3f_3 , s^4f_4	$\mathcal{Q}(s)$, $\mathcal{Q}_3$, $\mathcal{Q}_4$	$Q(s)$, Q_3 , Q_4	the integer inverse $Q(y)$ of §6; $Q(x, c)$ of Appendix A
continuum saddle; its bound	s ; $\bar{s}$	s ; S	the power sums $S_r(m)$
entropy; its coefficients	$\mathcal{D}(\varepsilon)$; d_r	$D(\varepsilon)$; d_r	$D_j(m)$, D_c ; the inverse corrections d_0 , d_1 ; d in (6.6)
hierarchy approximant	$\mathcal{V}_{r-1}(m, N)$	$M_{r-1}(m, N)$	M
$\log F_m(e^{-t})$	$\Lambda_m(t)$	$L_m(t)$	$L_r(c)$
geometric variables	ξ_j	Z_j	Z_m
Edgeworth coefficients; $\sum rk_r$	$\mathcal{C}_j(t)$; R	$C_j(t)$; R	C_v , C_0 in the proof of Theorem 7.3; $R_m(N)$
modulus of the j th factor	$\varrho_j(\theta)$	$R_j(\theta)$	$R_m(N)$; note $\prod_j \varrho_j = \rho_m$ at $\tau = t$
a large constant in Lemma 16.2	K_*	K	the truncation order K
Euler–Maclaurin terms	$A_k(s)$	$A_k(s)$	A_m of Lemma 2.1; $\mathcal{A}_m(t)$
local coordinate, $z = s\zeta$	ζ , $\phi_s(\zeta)$	w , $\phi_s(w)$	$w = W(ce^2L)$ in (6.2)
formal variable $m^{-1/2}$	$\hbar$	u	the pgf variable u of §7
standard normal variable	g	Y	the target value Y below
Gaussian coefficient operator	$\mathcal{G}(\hbar, s)$	$G(u, s)$	$G_q(x)$ of (6.5)
inverse input	Z , $\eta_* = m^2/Z$	Z , η	$\eta = m^3/N$ of Theorem 5.1; Z_m
inverse factor; corrections	$\mathcal{R}(\eta_*)$; ω , ω_k	$R(\eta)$; v , v_k	$R_m(N)$; v is kept for the contour parameter $t = s(1 + iv)/m$
sample value; threshold	Y ; $T_m(Y)$	Y ; $T_m(Y)$	Part I writes y ; $T = m(m-1)/2$ in (7.6)
surrogate; its root	$\mathcal{L}_J(X)$; X_J	$\mathcal{L}_J(X)$; X_J	$L_r(c)$
slice parameters	$\lambda, \gamma, \beta = \gamma - 2$, $b_\lambda = 2 + \log \lambda$	λ, γ, β , b	λ_r , $\gamma_r(m)$, $\beta_j(m)$, $b = \log q$; b_λ is the analogue of Part I's b_c
$\log Y$	L	L	the same as Part I's $L = \log y$

13 The research question and its precise scope

13.1 Why this problem was selected

The finite Euler product

$$F_m(q) = \prod_{j=1}^m \frac{1}{1 - q^j}, \quad p_m(N) = [q^N]F_m(q) \quad (13.1)$$

occurs in several useful OEIS sequences. A238016 records $p_m(m^k)$ as a two-dimensional array; its quadratic and cubic columns are A206226 and A238608. A097356 is $p_{\lfloor \sqrt{N} \rfloor}(N)$ [1, 2, 13]. These are regular partition families with a common generating function, not isolated numerical coincidences.

Topic	Part I	Part II (manuscript 09 of batch 98)
Uniform range	centered expansion for $N \geq c_0 m^3$ (Theorem 3.1); dilute expansion for $N \geq m^3/\eta_0$ (Theorem 5.1)	one logarithmic expansion for $N \geq am^2$, every fixed $a > 0$, no upper bound on N , with analytic coefficient functions of $\varepsilon = m^2/X$ (Theorem 14.1)
Minor arcs	near and far arcs for $\tau \leq C_0 m^{-2}$ (Lemma 3.2)	near arc for $t \leq \bar{s}/m$; far-arc integral $O(t(1 + \log m)e^{-cm})$ by reserving the factor $j = 1$ (Lemmas 16.1 and 16.2)
Volume law	$p_m(N) \sim V_m(N)$ iff $N/m^3 \rightarrow \infty$; limit $e^{1/(4c)}$ (Theorem 2.3, Corollary 4.1)	$p_m(N) \sim V_m(X)$ iff $X/m^{5/2} \rightarrow \infty$; limit $\exp(-1/(72\lambda^2))$ (Theorem 14.2)
Hierarchy	—	transition scales $X \asymp m^{2+1/(2r)}$ with limits $\exp(\mathbf{d}_r/\lambda^{2r})$, rational $\mathbf{d}_r$, infinitely many nonzero, $\mathbf{d}_1, \dots, \mathbf{d}_6 < 0$ (Theorem 14.3)
Exact saddle	—	Edgeworth expansion at the exact saddle, uniform for $N \geq am^2$ (Theorem 15.2)
Inverse	cubic Lambert chart, exponential-sector and A238000 inverses (Theorem 6.1, (6.8), (6.11))	analytic chart $\eta_* = \varepsilon e^{-\mathcal{D}(\varepsilon)}$ on $[0, e^2)$ to every fixed order (Theorem 14.4); threshold transfer (Proposition 22.2); Lambert inverse on power-law slices (22.12)
Coefficients	$L_1, \dots, L_5$; A238608 through m^{-5}	series of $\mathcal{D}$, E_0 , E_1 , $\mathcal{R}$, ω ; at $N = m^3$ only $-61/(288m)$
Repeated parts	Poisson law and its correction (Theorems 7.2, 7.3)	—
Checks	258 exact samples, $m \leq 60$, 90 digits	46 exact samples, $m \leq 96$, 55 digits

Table 1: Part II against Part I.

Earlier result	Manuscript 09's route
Theorem 3.1	Theorem 14.1 on $\varepsilon \leq 1/(c_0 m)$, in logarithmic form: continuum saddle, scaled coordinate, reserved factor; the two expansions agree after re-expansion, checked at order X^{-2} by (19.10)
(3.9) in Lemma 3.2	Lemma 16.1, the same proof under $t \leq \bar{s}/m$
(3.10) in Lemma 3.2	not reproved: replaced by Lemma 16.2, a different bound that survives $\tau \asymp 1/m$
Corollary 4.1	(21.4), from Theorem 14.1
(4.11), first correction	(21.5), also checked symbolically
(3.5), the term $j = 1$ of (3.8)	(19.10)
the notes on <code>p0:thm:staircase</code> , <code>p6:thm:gamma</code> , <code>p0:thm:perturbed-inversion</code> in §6	the same apparatus: Proposition 22.2 and (22.12)
<code>srp:eq:u-one</code> , <code>srp:eq:constants-one</code> (neighbouring report)	(21.2), the Szekeres leading formula
<code>srp:thm:allorders</code> (neighbouring report)	Theorem 14.1 on compact α -ranges, in other coordinates

Table 2: Results proved again by manuscript 09, and where Part II records each route. Manuscript 09 did not know these labels; the write identified them.

The selected repository reports are *Three Exceptions Before Every Square* [11] and *The Cubic Boundary for Restricted Partitions* (Part I of this report). The former develops expansions when N/m^2 stays in a compact subset of $(0, \infty)$; the latter develops centered expansions for $N \geq c_0 m^3$. The latter’s scope statement explicitly says that it does not uniformly match the quadratic regime and identifies this as the zero-saddle end of its neighbor’s research question. We address that end: N/m^2 may tend to infinity arbitrarily quickly. We do not address the opposite end $N/m^2 \rightarrow 0$ uniformly.

[write] **Added 5 October 2026, batch 98: where the cited statements are.** Manuscript 09 read the two reports’ READMEs only, so it cites no section or label of either. The “scope statement” is the README’s section “Relation to the repository”, printed in the article as Part I’s provenance note (§1.4); the centered expansion for $N \geq c_0 m^3$ is Theorem 3.1; the neighbour’s question is §14.6 of the report `a097356-sqrt-restricted-partitions`, “Uniform limits as the saddle moves to zero or infinity” (in its Section 14, label `srp:sec:future`), whose end $\alpha = N/m^2 \rightarrow \infty$ is addressed here. The same task is Part I’s own research question 2 (§10), “A uniform bridge below the cubic scale”; §20 answers it (see the dated note there).

The repository also identifies a Lean development of the finite-product identity in its Fabius-function project. We use the elementary identity (13.1), proved directly below, and do not import or claim any machine-checked asymptotic result.

[write] **Added 5 October 2026, batch 98.** The Lean file is `PartitionBoundedParts.lean` of the Fabius project, described with its full path in §1.3; it declares `boundedCount` and proves `hasSum_boundedCount_mul_pow`. Nothing in Part II is formalized, and its place beside that development gives it no formal status.

13.2 Historical boundary and novelty discipline

Szekeres’ leading formula already joins the main restricted-partition regimes. Canfield’s primary paper gives a proof and precise uniform ranges [5]. Consequently, a leading matching formula, or a consequence obtained merely by expanding that formula, should not be presented as a newly resolved historical conjecture. The posted leading law for A238608 is also already recorded in OEIS [2].

This report develops a uniform *all-fixed-orders* expansion with an explicit treatment of arbitrarily small saddles; a centered hierarchy of successive approximation boundaries; a rational coefficient algorithm; and a corresponding inverse chart. These are extensions of the two inspected repository reports. Their organization and explicit formulas were not located in the bounded source search, but we do not claim that every consequence or coefficient is absent from the broader partition literature. The saddle method, Euler–Maclaurin summation, Lagrange inversion and Lambert- W inversion are classical tools.

There is another relevant distinction. For fixed m , $p_m(N)$ is a quasipolynomial, and its root-of-unity components are often called Sylvester waves [7, 8]. Here m grows. We bound the noncentral arcs collectively; we do not construct the individual waves or claim optimal exponential accuracy.

13.3 Notation and meaning of an all-order statement

Throughout the main results, $m \rightarrow \infty$, m, N are positive integers, and $a > 0$ is fixed. Put

$$X = N + \frac{m(m+1)}{4}, \quad \varepsilon = \frac{m^2}{X}, \quad V_m(X) = \frac{X^{m-1}}{m! \Gamma(m)}. \quad (13.2)$$

The centered variable need not be integral. If $N \geq am^2$, then

$$0 < \varepsilon < \varepsilon_a := \frac{1}{a + 1/4} < 4. \quad (13.3)$$

An error $O_{a,J}(m^{-J-1})$ has one constant valid for all such N , including superexponentially large N . The order J is fixed before $m \rightarrow \infty$. No uniformity in a growing J is asserted.

14 Main results

Define the regularized logarithm and its mean by

$$h(z) = \log \frac{z}{2 \sinh(z/2)}, \quad h(0) = 0, \quad H(s) = \int_0^1 h(sx) dx. \quad (14.1)$$

On the positive real axis the logarithm is real. Near zero the analytic branch is fixed by $h(0) = 0$. Write

$$B(s) = 1 - sH'(s) = 1 - h(s) + H(s), \quad Q(s) = 1 + s^2H''(s). \quad (14.2)$$

For $0 < \varepsilon < 4$, let $s = s(\varepsilon) > 0$ be the unique solution of

$$\frac{1}{\varepsilon} = \frac{1}{s} - H'(s), \quad \text{equivalently} \quad s = \varepsilon B(s). \quad (14.3)$$

Existence, uniqueness and regularity will be proved in Section 17. Set

$$\mathcal{D}(\varepsilon) = \frac{s}{\varepsilon} - 1 - \log \frac{s}{\varepsilon} + H(s), \quad (14.4)$$

$$E_0(\varepsilon) = \frac{h(s)}{2} + \log \frac{s}{\varepsilon} - \frac{1}{2} \log Q(s). \quad (14.5)$$

At zero both functions are defined by continuity and equal zero.

Theorem 14.1 (Uniform centered expansion). *There are real-analytic functions E_j on $[0, 4)$, $j \geq 1$, such that, for every $a > 0$ and every fixed integer $J \geq 0$,*

$$\log \frac{p_m(N)}{V_m(X)} = m\mathcal{D}(\varepsilon) + E_0(\varepsilon) + \sum_{j=1}^J \frac{E_j(\varepsilon)}{m^j} + O_{a,J}(m^{-J-1}), \quad N \geq am^2. \quad (14.6)$$

Every E_j has an even analytic germ at zero and satisfies $E_j(0) = 0$. The same properties hold for $\mathcal{D}$ and E_0 . A finite Gaussian-coefficient algorithm computes each E_j from h, H and Bernoulli numbers.

Here “analytic on $[0, 4)$ ” means analytic in a neighborhood of each point of that half-open real interval, not one unspecified complex disk of radius four. Exponentiating (14.6) gives a relative error $1 + O_{a,J}(m^{-J-1})$ after retaining the displayed exponential.

[write] **Added 5 October 2026, batch 98: relation to Part I.** On the range $N \geq c_0 m^3$ of Theorem 3.1 one has $\varepsilon \leq 1/(c_0 m)$, and both theorems determine $\log(p_m(N)/V_m(X))$ to $O(m^{-K-1})$ uniformly there (X is Part I’s M). They are therefore two expansions of one quantity and agree after re-expansion in ε ; the first-wave check (19.10) is this agreement at order X^{-2} , against Part I’s a_1 term (3.5). Theorem 14.1 is thus a second route to Theorem 3.1, in logarithmic form, and extends it from $N \geq c_0 m^3$ to $N \geq am^2$; manuscript 09 does not carry out the re-expansion beyond order X^{-2} . On compact α -ranges it likewise covers, in other

coordinates, the bounded-offset expansion `srp:thm:allorders` of the report `a097356-sqrt-restricted-partitions`, which uses the same regularized logarithm h . (Dated note, 7 October 2026: not the same function. That report's regularized logarithm is $\log(z/(1 - e^{-z}))$ (its Proposition 4.1, `srp:prop:product`), and $\log(z/(1 - e^{-z})) = h(z) + z/2$: the two differ by the linear term $z/2$. This note corrects only that clause.) The proof in §18 is condensed; Remark 18.1 completes it.

Theorem 14.2 (Sharp centered boundary). *For every sequence with $N \geq am^2$,*

$$p_m(N) \sim V_m(X) \iff X/m^{5/2} \longrightarrow \infty. \quad (14.7)$$

At the boundary, for each fixed $\lambda > 0$,

$$X \sim \lambda m^{5/2} \implies \frac{p_m(N)}{V_m(X)} \longrightarrow \exp\left(-\frac{1}{72\lambda^2}\right). \quad (14.8)$$

This is not a change to the previously studied *uncentered* cubic boundary: $V_m(N)$ and $V_m(X)$ are different approximations.

Expand the entropy at zero as $\mathcal{D}(\varepsilon) = \sum_{r \geq 1} \mathbf{d}_r \varepsilon^{2r}$.

Theorem 14.3 (Crossover hierarchy). *The coefficients are rational and satisfy*

$$\mathbf{d}_r = -\frac{[z^{2r}]B(z)^{2r+1}}{2r(2r+1)}. \quad (14.9)$$

For a fixed r with $\mathbf{d}_r \neq 0$, define

$$\mathcal{V}_{r-1}(m, N) = V_m(X) \exp\left(m \sum_{j=1}^{r-1} \mathbf{d}_j (m^2/X)^{2j}\right). \quad (14.10)$$

Then

$$X \sim \lambda m^{2+1/(2r)} \implies \frac{p_m(N)}{\mathcal{V}_{r-1}(m, N)} \longrightarrow \exp(\mathbf{d}_r/\lambda^{2r}). \quad (14.11)$$

Infinitely many $\mathbf{d}_r$ are nonzero. In particular there are infinitely many distinct transition exponents accumulating at 2 from above. The first six $\mathbf{d}_r$ are negative and are listed in Table 4. Nonvanishing and negativity for every r are not claimed.

Theorem 14.4 (Uniform analytic inverse chart). *Let $Y = p_m(N)$ with $N \geq am^2$, and define*

$$Z = (Ym!\Gamma(m))^{1/(m-1)}, \quad \eta_\star = m^2/Z. \quad (14.12)$$

For $0 \leq \eta_\star < e^2$ define ε by

$$\eta_\star = \varepsilon e^{-\mathcal{D}(\varepsilon)}, \quad \mathcal{R}(\eta_\star) = e^{-\mathcal{D}(\varepsilon)}. \quad (14.13)$$

The map is uniquely invertible and analytic, including at zero. With

$$\omega(\eta_\star) = \frac{-\mathcal{D}(\varepsilon) - E_0(\varepsilon)}{1 - \varepsilon \mathcal{D}'(\varepsilon)}, \quad (14.14)$$

we have, uniformly in the stated sample-value regime,

$$X = Z\mathcal{R}(\eta_\star) \exp\left(\frac{\omega(\eta_\star)}{m} + O_a(m^{-2})\right). \quad (14.15)$$

There is an analytic inverse expansion to every fixed order in $1/m$. These are smooth asymptotic charts, not exact integer-rounding rules.

15 Exact generating functions and a finite-saddle expansion

For each possible part j , its multiplicity contributes $1 + q^j + q^{2j} + \dots$. Multiplying these choices proves (13.1) coefficient by coefficient. It also proves the exact recurrence

$$p_m(N) = p_{m-1}(N) + p_m(N - m), \quad (15.1)$$

where negative arguments have value zero, $p_m(0) = 1$, and $p_0(N) = 0$ for $N > 0$.

For $t > 0$ let

$$\Lambda_m(t) = \log F_m(e^{-t}) = - \sum_{j=1}^m \log(1 - e^{-jt}). \quad (15.2)$$

Define positive cumulants

$$\kappa_r(t) = \sum_{j=1}^m j^r \sum_{\ell \geq 1} \ell^{r-1} e^{-j\ell t}, \quad r \geq 1. \quad (15.3)$$

Then $(-1)^r \Lambda_m^{(r)}(t) = \kappa_r(t)$ and $\kappa_1(t) = \sum j/(e^{jt} - 1)$. The function κ_1 decreases continuously from infinity to zero, so there is a unique exact saddle $t = t_{m,N}$ satisfying

$$\kappa_1(t) = N. \quad (15.4)$$

Since $j/(e^{jt} - 1) \leq 1/t$, this saddle satisfies $t \leq m/N \leq 1/(am)$.

There is a useful probabilistic interpretation. Let independent ξ_j have the geometric distribution $\Pr(\xi_j = k) = (1 - e^{-jt})e^{-jtk}$, $k \geq 0$. Then

$$p_m(N) = e^{Nt + \Lambda_m(t)} \Pr\left(\sum_{j=1}^m j\xi_j = N\right). \quad (15.5)$$

At the exact saddle the random sum has mean N . Its centered characteristic function gives

$$p_m(N) = \frac{e^{Nt + \Lambda_m(t)}}{2\pi} \int_{-\pi}^{\pi} \exp\{\Lambda_m(t - i\theta) - \Lambda_m(t) - iN\theta\} d\theta. \quad (15.6)$$

Lemma 15.1 (Uniform cumulant scale). *For fixed $\bar{s} > 0$ and fixed $r \geq 1$, there are positive constants $c_{r,\bar{s}}, C_{r,\bar{s}}$ such that, whenever $0 < t \leq \bar{s}/m$,*

$$c_{r,\bar{s}} m t^{-r} \leq \kappa_r(t) \leq C_{r,\bar{s}} m t^{-r}. \quad (15.7)$$

Proof. For $y > 0$, the function $y^r \sum_{\ell \geq 1} \ell^{r-1} e^{-y\ell}$ is positive and continuous. As $y \downarrow 0$ it tends to $(r-1)!$, by a Riemann-sum comparison with $\int_0^\infty x^{r-1} e^{-x} dx$. Thus it is bounded above and away from zero on $[0, \bar{s}]$. Apply this observation to $y = jt$ in each summand of (15.3). $\square$

Theorem 15.2 (Exact-saddle expansion). *For each fixed $J \geq 0$, uniformly for $N \geq am^2$,*

$$p_m(N) = \frac{e^{Nt + \Lambda_m(t)}}{\sqrt{2\pi\kappa_2(t)}} \left(\sum_{j=0}^J \mathcal{C}_j(t) + O_{a,J}(m^{-J-1}) \right), \quad (15.8)$$

where $\mathcal{C}_0 = 1$, $\mathcal{C}_j = O_{a,j}(m^{-j})$, and

$$\mathcal{C}_j = \sum_{\substack{k_3, k_4, \dots \geq 0 \\ \sum_{r \geq 3} (r-2)k_r = 2j}} (-1)^{R/2} (R-1)!! \prod_{r \geq 3} \frac{1}{k_r!} \left(\frac{\kappa_r}{r! \kappa_2^{r/2}} \right)^{k_r}, \quad R = \sum_{r \geq 3} r k_r. \quad (15.9)$$

Only finitely many k_r can be nonzero, and R is even under the constraint. In particular,

$$\mathcal{C}_1 = \frac{\kappa_4}{8\kappa_2^2} - \frac{5\kappa_3^2}{24\kappa_2^3}. \quad (15.10)$$

The nontrivial uniformity issue in this theorem is not the formal Gaussian expansion: it is showing that the noncentral contour is negligible even when t is extremely small. We do this next, then complete the proof.

16 A minor-arc estimate that survives an arbitrarily small saddle

For any $0 < t \leq \bar{s}/m$, the modulus of the j th characteristic factor is

$$\varrho_j(\theta) = \frac{1 - e^{-jt}}{|1 - e^{-jt + i j \theta}|} = \left(1 + \frac{\sin^2(j\theta/2)}{\sinh^2(jt/2)} \right)^{-1/2}. \quad (16.1)$$

This identity avoids cancellation and displays both relevant scales.

Lemma 16.1 (Small-angle product bound). *There is $c_{\bar{s}} > 0$ such that*

$$\prod_{j=1}^m \varrho_j(\theta) \leq \left(1 + c_{\bar{s}}(\theta/t)^2 \right)^{-m/2}, \quad |\theta| \leq 1/m. \quad (16.2)$$

Proof. On $|u| \leq 1/2$, $|\sin u| \geq c|u|$ for an absolute $c > 0$. On $0 \leq v \leq \bar{s}/2$, $\sinh v \leq C_{\bar{s}}v$. Apply these bounds to $u = j\theta/2$ and $v = jt/2$ in (16.1); the factor j cancels. Multiply the m resulting bounds. $\square$

[write] **Added 5 October 2026, batch 98: Part I's near-arc bound.** $\prod_j \varrho_j(\theta)$ is Part I's $\rho_m(\theta)$ at $\tau = t$, and (16.1) is the factor identity in the proof of Lemma 3.2, since $4e^{-x}/(1-e^{-x})^2 = 1/\sinh^2(x/2)$. Lemma 16.1 is therefore (3.9) with the same proof, under the weaker hypothesis $t \leq \bar{s}/m$ in place of $\tau \leq C_0 m^{-2}$. Part I's far-arc bound (3.10), $(Cm\tau)^{\delta m}$, needs $Cm\tau < 1$ and gives nothing when $\tau \asymp 1/m$, which is the quadratic scale; Lemma 16.2 replaces it there.

Lemma 16.2 (Macroscopic-angle bound). *For fixed $\bar{s} > 0$, there are constants $c_{\bar{s}}, C_{\bar{s}} > 0$ such that, for all sufficiently large m ,*

$$\int_{1/m \leq |\theta| \leq \pi} \prod_{j=1}^m \varrho_j(\theta) d\theta \leq C_{\bar{s}} t (1 + \log m) e^{-c_{\bar{s}} m}. \quad (16.3)$$

Proof. First we show

$$\sum_{j=2}^m \sin^2(j\theta/2) \geq cm, \quad 1/m \leq |\theta| \leq \pi, \quad (16.4)$$

for one absolute $c > 0$. Fix a sufficiently large constant K_* . For $1/m \leq |\theta| \leq K_*/m$, set $u = m|\theta|$. The Riemann sums $m^{-1} \sum_{j=2}^m \sin^2(uj/(2m))$ converge uniformly on $u \in [1, K_*]$ to $\int_0^1 \sin^2(ux/2) dx$. The continuous limiting function is strictly positive on that compact interval. For $K_*/m \leq |\theta| \leq \pi$, the geometric-sum estimate gives $|\sum_{j=1}^m e^{ij\theta}| \leq 1/|\sin(\theta/2)| \leq \pi/|\theta|$. Using $\sin^2 v = (1 - \cos 2v)/2$ proves (16.4) when K_* is large, after absorbing the deleted $j = 1$ term.

Since $jt \leq \bar{s}$, (16.1) implies $\log \varrho_j(\theta) \leq -c'_s \sin^2(j\theta/2)$. Spend the factors $j = 2, \dots, m$ on (16.4); their product is at most $e^{-c_s m}$. Keep the first factor separately:

$$\varrho_1(\theta) \leq \frac{\sinh(t/2)}{|\sin(\theta/2)|} \leq \frac{C_s t}{|\sin(\theta/2)|}. \quad (16.5)$$

Integrating its reciprocal-sine bound from $1/m$ to π gives $O(t(1 + \log m))$, proving (16.3). $\square$

Remark 16.3 (Why the factor t matters). The principal integral has size $\kappa_2^{-1/2} \asymp t/\sqrt{m}$. An absolute tail estimate $O(e^{-cm})$ would not establish a relative error uniformly in unbounded N : t could be smaller than e^{-m^2} . The reserved $j = 1$ factor in Lemma 16.2 supplies precisely the missing t . The proof does not require an upper growth condition on N .

16.1 Completion of the finite-saddle proof

Choose a fixed small $\delta > 0$. The central window $|\theta| \leq tm^{-1/2+\delta}$ lies inside $|\theta| \leq 1/m$ for large m . By Lemma 16.1 and the substitution $x = \theta/t$, the integral between these windows is

$$O\left(tm^{-1/2}e^{-cm^{2\delta}}\right). \quad (16.6)$$

Indeed, on $|x| \leq 1$ the bound in (16.2) is Gaussian; on $|x| \geq 1$ its integral is exponentially small in m . Lemma 16.2 handles the remaining contour. Both bounds are smaller than tm^{-A} for every fixed A .

Inside the central window use $y = \theta\sqrt{\kappa_2}$. Exact centering cancels the linear term, and the exponent is

$$-\frac{y^2}{2} + \sum_{r \geq 3} \frac{\kappa_r (\mathrm{i}y)^r}{r! \kappa_2^{r/2}}. \quad (16.7)$$

The functions involved are analytic for $|\theta| < ct$, for any fixed $c < 1$. Cauchy estimates and Lemma 15.1 bound the coefficient of degree r by a constant times $m^{1-r/2}$. For a prescribed J , take δ sufficiently small, Taylor-expand to a finite degree exceeding $2J + 4$, and expand the exponential to the corresponding finite weight. The remainder is bounded by m^{-J-1} times an integrable polynomial in $|y|$ times e^{-cy^2} . Terms of odd total degree integrate to zero on the symmetric window; extending the finitely many Gaussian-polynomial integrals to the full line costs an exponentially small error. The Gaussian moments are $\mathbb{E}g^{2k} = (2k-1)!!$ for $g \sim N(0, 1)$. Grouping by $\sum (r-2)k_r = 2j$ gives (15.9), with its sign $\mathrm{i}^R = (-1)^{R/2}$. These estimates prove Theorem 15.2, including its fixed-order remainder.

[write] **Added 5 October 2026, batch 98: a direct derivative bound.** The Cauchy estimate can be replaced by a bound valid for every real θ : for $\operatorname{Re} t' > 0$ the series $\Lambda_m^{(r)}(t') = (-1)^r \sum_j j^r \sum_\ell \ell^{r-1} e^{-jt'\ell}$ converges absolutely, and $|e^{-j(t-i\theta)\ell}| = e^{-jt\ell}$, so $|\Lambda_m^{(r)}(t-i\theta)| \leq \kappa_r(t)$. The Taylor remainder of order $R' + 1$ along the segment from t to $t - i\theta$ is therefore at most $|\theta|^{R'+1} \kappa_{R'+1}(t)/(R' + 1)!$, which is $O(m(|y|/\sqrt{m})^{R'+1})$ by Lemma 15.1.

17 Centering and the continuum saddle

The elementary identity

$$\frac{1}{1 - e^{-z}} = z^{-1} e^{z/2} e^{h(z)} \quad (17.1)$$

gives the exact formula

$$Nt + \Lambda_m(t) = Xt - m \log t - \log(m!) + \sum_{j=1}^m h(jt). \quad (17.2)$$

Thus the displacement $m(m+1)/4$ removes the entire linear term in the logarithm of the product. What remains, h , is even at the origin. This is why centering changes the first approximation boundary.

The identity $(sH(s))' = h(s)$ proves the second expression for B in (14.2). Differentiating under the integral also gives

$$\mu(s) := \frac{1}{s} - H'(s) = \int_0^1 \frac{x}{2} \coth(sx/2) dx = \frac{1}{4} + \int_0^1 \frac{x}{e^{sx} - 1} dx, \quad (17.3)$$

$$-\mu'(s) = \int_0^1 \frac{x^2 e^{sx}}{(e^{sx} - 1)^2} dx > 0. \quad (17.4)$$

In particular $\mathcal{Q}(s) = -s^2 \mu'(s) > 0$.

Lemma 17.1 (Analytic saddle parametrization). *The map $s \mapsto \varepsilon = 1/\mu(s) = s/B(s)$ is a strictly increasing bijection from $(0, \infty)$ onto $(0, 4)$. Its inverse $s(\varepsilon)$ is real analytic there, extends analytically at zero, and is odd near zero. Moreover $B(s) > 1$ for every $s > 0$, $\mathcal{Q}(0) = 1$, and $s(\varepsilon) = \varepsilon + O(\varepsilon^3)$.*

Proof. Equations (17.3)–(17.4) show strict monotonicity, $\mu(s) \rightarrow \infty$ at zero and $\mu(s) \rightarrow 1/4$ at infinity. The nonzero derivative gives local analytic inverses. Near zero h, H are even analytic functions, $B(0) = 1$, and the analytic implicit-function theorem applies to $s - \varepsilon B(s) = 0$. Symmetry and uniqueness make s odd. Finally $B(s) = \int_0^1 (sx/2) \coth(sx/2) dx > 1$ because $u \coth u > 1$ for $u > 0$. The other limits follow directly. $\square$

The continuum saddle for the main parameter ε is $t = s/m$. For fixed $a > 0$, s ranges in the compact interval $[0, s(\varepsilon_a)]$. Thus the contour estimates of Section 16 apply to this saddle as well. It is not exactly $t_{m,N}$, but its discrepancy is absorbed by the analytic endpoint amplitude below; no claim that it is an exact mean is made.

18 Uniform Euler–Maclaurin analysis and a complete coefficient rule

For every fixed truncation order, Euler–Maclaurin summation gives

$$\sum_{j=1}^m h(js/m) \sim mH(s) + \frac{h(s)}{2} + \sum_{k \geq 1} \frac{A_k(s)}{m^{2k-1}}, \quad A_k(s) = \frac{B_{2k}}{(2k)!} s^{2k-1} h^{(2k-1)}(s). \quad (18.1)$$

The Bernoulli numbers have $B_2 = 1/6$. The terms at zero in the derivative differences vanish because h is even. This is a uniform fixed-order expansion on every compact s -interval, including zero, and

on a sufficiently small complex neighborhood of that interval along the local saddle contour. To justify any requested remainder, apply Euler–Maclaurin with more terms than needed: the periodic Bernoulli remainder is bounded by the corresponding derivative supremum, which is uniform on that compact neighborhood. Thus no formal interchange with a singular endpoint is being used.

18.1 A nonsingular local coordinate

Use $z = s\zeta$ and define

$$\phi_s(\zeta) = B(s)\zeta - \log \zeta + H(s\zeta). \quad (18.2)$$

It satisfies

$$\phi'_s(1) = 0, \quad \phi''_s(1) = \mathcal{Q}(s) > 0. \quad (18.3)$$

This coordinate is important: the derivatives of $z/\varepsilon - \log z + H(z)$ diverge as $s \rightarrow 0$, whereas the derivatives of ϕ_s remain uniformly bounded near $\zeta = 1$. Moreover $\mathcal{Q}(s)$ is uniformly bounded above and away from zero on the compact range relevant to $N \geq am^2$.

The contour may be parametrized by $t = s(1 + iv)/m$; reversing its orientation or replacing v by $-v$ does not change the real result. With $v = y/\sqrt{m\mathcal{Q}(s)}$, the normalized saddle integral is Gaussian. Its leading term, using (17.2) and (18.1), is

$$\frac{e^{m(B(s)+H(s))}(m/s)^m}{m!} \frac{s}{m\sqrt{2\pi m\mathcal{Q}(s)}} e^{h(s)/2}. \quad (18.4)$$

Stirling's expansion of $\Gamma(m)$ shows that the ratio of (18.4) to $V_m(X)$ is $\exp(m\mathcal{D}(\varepsilon) + E_0(\varepsilon))$ times the Stirling correction series. This proves the displayed forms of $\mathcal{D}$ and E_0 .

18.2 An explicit all-order operator

The following formal prescription fixes every higher coefficient without an unspecified appeal to “standard saddle coefficients.” Let $\hbar$ be a formal variable and g a standard normal variable, and put

$$\zeta = 1 + \frac{i\hbar g}{\sqrt{\mathcal{Q}(s)}}. \quad (18.5)$$

Define $\mathcal{G}(\hbar, s)$ by formally expanding

$$\begin{aligned} \mathcal{G}(\hbar, s) = \mathbb{E} \exp \left\{ \frac{\phi_s(\zeta) - \phi_s(1) + \hbar^2 g^2/2}{\hbar^2} + \frac{h(s\zeta) - h(s)}{2} \right. \\ \left. + \sum_{k \geq 1} A_k(s\zeta) \hbar^{4k-2} \right\}. \end{aligned} \quad (18.6)$$

At every requested power of $\hbar$, this means finite Taylor expansion, followed by replacement of Gaussian monomials by their moments. The apparent negative power of $\hbar$ causes no problem: the saddle and quadratic terms cancel in the numerator. Odd coefficients vanish after Gaussian averaging. In particular $\mathcal{G}(0, s) = 1$.

For $j \geq 1$ set

$$E_j(\varepsilon) = [\hbar^{2j}] \left\{ \log \mathcal{G}(\hbar, s(\varepsilon)) + \sum_{k \geq 1} \frac{B_{2k}}{2k(2k-1)} \hbar^{4k-2} \right\}. \quad (18.7)$$

The second sum is the logarithmic Stirling series for $\Gamma(m)$, with $\hbar = m^{-1/2}$. Formula (18.7) is both a definition and an effective coefficient algorithm: only finitely many derivatives and Gaussian moments occur for any fixed j .

18.3 Proof of the uniform theorem

On the compact s -range, the derivatives in (18.6) and a fixed number of Euler–Maclaurin remainders have uniform bounds. The local Gaussian argument used for Theorem 15.2 therefore gives a uniform finite expansion, with error $O(m^{-J-1})$ after taking sufficiently many terms. The contour outside that window is negligible relative to its leading size $s/m^{3/2}$ by Lemmas 16.1–16.2, uniformly even as $s \rightarrow 0$. The leading amplitude in (18.4), after normalization, stays positive and bounded away from zero. Thus the logarithm of the normalized expansion is legitimate for large m and has coefficients (18.7).

All operations defining those coefficients are analytic on a local complex neighborhood of each real s in the compact range. The evenness of h, H, A_k and the oddness of $s(\varepsilon)$ give even germs in ε ; the normalized phase uses $B(s)$ and $H(s\zeta)$, which are even in s as well.

Finally, at $s = 0$ the phase is $\zeta - \log \zeta$, all A_k and h vanish, and its inverse-Laplace integral is

$$\frac{1}{2\pi i} \int_{1-i\infty}^{1+i\infty} e^{m\zeta} \zeta^{-m} d\zeta = \frac{m^{m-1}}{\Gamma(m)}. \quad (18.8)$$

It follows that $\mathcal{G}(\hbar, 0)$ is exactly the reciprocal Stirling series appearing in (18.7); their logarithms cancel coefficient by coefficient. Hence $E_j(0) = 0$ for all j . This establishes Theorem 14.1. The evaluation at $s = 0$ is an identity of asymptotic coefficient systems, not a statement that a finite partition problem has $\varepsilon = 0$.

Remark 18.1 ([write] Completion of the proof of Theorem 14.1; added 5 October 2026, batch 98). The proof above is condensed: it does not say on which circle Cauchy’s formula is applied, why the continuum saddle may replace the exact one, or on which complex domain Euler–Maclaurin is uniform. The following fills these steps; it uses only Lemmas 16.1 and 16.2, Lemma 17.1 and the displays of this section. Fix $a > 0$ and put $\bar{s} = s(\varepsilon_a)$. For $N \geq am^2$, (13.3) and the monotonicity in Lemma 17.1 give $0 < s < \bar{s}$, where $s = s(\varepsilon)$.

- (i) *The circle.* Apply Cauchy’s formula on $|q| = e^{-\tau}$ with $\tau = s/m$, the continuum saddle, so $\tau \leq \bar{s}/m$ and both lemmas of §16 hold with $t = \tau$. With $t = \tau + i\theta$,

$$p_m(N) = \frac{e^{N\tau} F_m(e^{-\tau})}{2\pi} \int_{-\pi}^{\pi} \frac{e^{iN\theta} F_m(e^{-\tau-i\theta})}{F_m(e^{-\tau})} d\theta,$$

and the modulus of the integrand is $\prod_j \varrho_j(\theta)$ with $t = \tau$. Nothing in the argument uses the exact saddle $t_{m,N}$.

- (ii) *The exponent in the window.* On the window $|\theta| \leq \tau m^{-1/2+\delta}$ write $t = s\zeta/m$, $\zeta = 1 + iv$, $|v| \leq m^{-1/2+\delta}$. Since $X = m^2/\varepsilon = m^2 B(s)/s$, one has $Xt = mB(s)\zeta$, and (17.2) becomes

$$Nt + \Lambda_m(t) = m\phi_s(\zeta) + \left(\sum_{j=1}^m h(js\zeta/m) - mH(s\zeta) \right) - m \log \frac{s}{m} - \log m!.$$

The function $z/(2 \sinh(z/2))$ has no zeros in the strip $|\operatorname{Im} z| < 2\pi$ (the zeros of $\sinh(z/2)$ other than $z = 0$ are $2\pi ik$, $k \neq 0$, and the quotient equals 1 at $z = 0$), so h is analytic and even there. Put $\rho_0 = \min(1/2, \pi/\bar{s})$. For $0 \leq s \leq \bar{s}$, $|\zeta - 1| \leq \rho_0$ and $0 \leq x \leq 1$ the point $xs\zeta$ has $|\operatorname{Im}(xs\zeta)| \leq \pi$, so $h, H, \phi_s, h(s\zeta)$ and $A_k(s\zeta)$ are analytic in ζ on $|\zeta - 1| \leq \rho_0$, with bounds on all their derivatives that depend only on $\bar{s}$ and the order. Euler–Maclaurin of order $2K$ for $f(x) = h(xs\zeta/m)$ on $[0, m]$ gives the bracket as $h(s\zeta)/2 + \sum_{k \leq K} A_k(s\zeta) m^{1-2k}$ plus

a remainder at most a constant times $m \cdot (\bar{s}(1 + \rho_0)/m)^{2K}$, uniformly: $f(0) = 0$, the odd derivatives of f vanish at $x = 0$ because h is even, $f^{(2k-1)}(m) = (s\zeta/m)^{2k-1}h^{(2k-1)}(s\zeta)$, and the remainder integral is bounded by $m \sup |f^{(2K)}|$ times a constant depending on K .

- (iii) *The Gaussian step.* Here $\phi'_s(1) = 0$ and $\phi''_s(1) = \mathcal{Q}(s) \in [\mathcal{Q}_-, \mathcal{Q}_+] \subset (0, \infty)$ for $0 \leq s \leq \bar{s}$. Put $\hbar = m^{-1/2}$ and $v = \hbar y / \sqrt{\mathcal{Q}(s)}$, so that $\zeta = 1 + i\hbar y / \sqrt{\mathcal{Q}(s)}$, the quadratic part of $m(\phi_s(\zeta) - \phi_s(1))$ is $-y^2/2$, and the window is $|y| \leq \sqrt{\mathcal{Q}(s)}m^\delta$. By Cauchy's estimates on $|\zeta - 1| \leq \rho_0$, the Taylor term of degree $r \geq 3$ of $m(\phi_s(\zeta) - \phi_s(1))$ is $O(\hbar^{r-2}|y|^r)$ uniformly in s . The continuum saddle is a critical point of ϕ_s only, not of the whole exponent: $(h(s\zeta) - h(s))/2$ has the linear term $sh'(s)i\hbar y / (2\sqrt{\mathcal{Q}(s)}) = O(\hbar|y|)$, which is not cancelled and is expanded together with the cubic and higher terms; it is the term $(h(s\zeta) - h(s))/2$ of (18.6), and it pairs with odd terms after Gaussian averaging. For $\delta < 1/6$ all these corrections, and the Euler–Maclaurin remainders, are $O(m^{-1/2+3\delta})$ uniformly on the window, so the exponent there is $-y^2/2 + o(1)$. Taylor expansion to a finite degree and expansion of the exponential to weight $2J + 2$ leave a remainder bounded by m^{-J-1} times a polynomial in $|y|$ times $e^{-y^2/2}$, provided δ is small in terms of J and the Euler–Maclaurin order satisfies $K \geq J + 1$. Extending the Gaussian-polynomial integrals from the window to $\mathbb{R}$ costs $O(e^{-cm^{2\delta}})$. With the constant terms $A_k(s)\hbar^{4k-2}$ kept inside $\mathcal{G}$, as in (18.6), and $d\theta = \tau\hbar dy / \sqrt{\mathcal{Q}(s)}$, the window contributes

$$\frac{\tau}{\sqrt{2\pi m \mathcal{Q}(s)}} \frac{e^{m(B(s)+H(s))+h(s)/2}(m/s)^m}{m!} (\mathcal{G}_{2J}(\hbar, s) + O(m^{-J-1}))$$

to $p_m(N)$, where $\mathcal{G}_{2J}$ is the truncation of (18.6) after $\hbar^{2J}$; with $\tau = s/m$ this is (18.4) times $\mathcal{G}_{2J} + O(m^{-J-1})$. Since $e^{N\tau} F_m(e^{-\tau}) = e^{m(B(s)+H(s))+h(s)/2+O(1/m)}(m/s)^m/m!$ by (ii) at $\zeta = 1$, the window integral, relative to $e^{N\tau} F_m(e^{-\tau})$, has size $\asymp \tau m^{-1/2}$.

- (iv) *The tails, relative to the window.* With $x = \theta/\tau$, Lemma 16.1 bounds the integral over $\tau m^{-1/2+\delta} \leq |\theta| \leq 1/m$ by

$$\tau \int_{m^{-1/2+\delta} \leq |x| \leq 1} e^{-c'mx^2} dx + \tau \int_{|x| \geq 1} (1 + cx^2)^{-m/2} dx = O(\tau e^{-c''m^{2\delta}}),$$

using $(1 + cx^2)^{-m/2} \leq e^{-c'mx^2}$ for $|x| \leq 1$ and $(1 + cx^2)^{-m/2} \leq (1 + c)^{-(m-2)/2}(1 + cx^2)^{-1}$ for $|x| \geq 1$. Lemma 16.2 bounds the integral over $1/m \leq |\theta| \leq \pi$ by $C\tau(1 + \log m)e^{-cm}$. Divided by the window size $\asymp \tau m^{-1/2}$, both are $O(m^{-A})$ for every A , uniformly in $0 < s \leq \bar{s}$: the factor τ cancels. This is where the reserved factor $j = 1$ of Lemma 16.2 is essential, since $\tau = s/m$ is unbounded below as $N/m^2 \rightarrow \infty$.

- (v) *Normalization.* At $\zeta = 1$ the exponent is $m(B(s) + H(s)) + h(s)/2 + \sum_k A_k(s)m^{1-2k} - m \log(s/m) - \log m!$, up to the Euler–Maclaurin remainder. Combining (iii) and (iv) with this and with Stirling's series for $\log \Gamma(m)$, exactly as for (18.4), gives $\log(p_m(N)/V_m(X)) = m\mathcal{D} + E_0 + \log \mathcal{G}_{2J}(\hbar, s) + \sum_k B_{2k}\hbar^{4k-2}/(2k(2k-1)) + O(m^{-J-1})$. Since $\mathcal{G}_{2J} = 1 + O(m^{-1})$ uniformly, its logarithm may be expanded, and the coefficients are (18.7).
- (vi) *Analyticity and parity.* Each $[\hbar^{2j}] \log \mathcal{G}$ is a polynomial in finitely many of $\phi_s^{(r)}(1) = (-1)^r(r-1)! + s^r H^{(r)}(s)$ ($r \geq 2$), $s^k h^{(k)}(s)$, the derivatives of $A_k(s\zeta)$ at $\zeta = 1$, and $\mathcal{Q}(s)^{-1}$: a monomial survives Gaussian averaging only if its total degree in g is even, so only integer powers of $\mathcal{Q}(s)^{-1}$ occur. These are analytic on a neighbourhood of $[0, \infty)$ (where $\mathcal{Q} > 0$) and even in s , because H , h and $z^{2k-1}h^{(2k-1)}(z)$ are even, so $s^r H^{(r)}(s)$ and $s^k h^{(k)}(s)$ are even. Composition with the odd analytic $s(\varepsilon)$ of Lemma 17.1 gives analytic E_j on $[0, 4)$ with even germs at

zero; $E_j(0) = 0$ is the gamma-integral argument above, made rigorous by the uniqueness of asymptotic expansions: the same Gaussian step applied to (18.8) gives $\mathcal{G}(\hbar, 0) e^m (2\pi m)^{-1/2}$ as the expansion of $m^{m-1}/\Gamma(m)$.

With (i)–(vi), the proof of Theorem 14.1 is complete at the level of detail of Part I’s proof of Theorem 3.1. It has not been independently refereed.

19 Explicit coefficients and internal consistency checks

The Bernoulli expansion is

$$h(z) = - \sum_{r \geq 1} \frac{B_{2r}}{2r(2r)!} z^{2r} = -\frac{z^2}{24} + \frac{z^4}{2880} - \frac{z^6}{181440} + \cdots, \quad (19.1)$$

$$H(z) = -\frac{z^2}{72} + \frac{z^4}{14400} - \frac{z^6}{1270080} + \cdots. \quad (19.2)$$

These series converge in a neighborhood of zero; no assertion about convergence of the $1/m$ expansion follows from that fact.

19.1 The entropy coefficients

The envelope derivative of (14.4) is especially simple:

$$\varepsilon \mathcal{D}'(\varepsilon) = 1 - \frac{s(\varepsilon)}{\varepsilon} = 1 - B(s(\varepsilon)). \quad (19.3)$$

Lagrange inversion applied to $s = \varepsilon B(s)$ yields

$$[\varepsilon^{2r+1}]s(\varepsilon) = \frac{[z^{2r}]B(z)^{2r+1}}{2r+1}. \quad (19.4)$$

Integrating (19.3) with $\mathcal{D}(0) = 0$ proves (14.9). For example,

$$s(\varepsilon) = \varepsilon + \frac{\varepsilon^3}{36} + \frac{41\varepsilon^5}{32400} + \frac{3749\varepsilon^7}{57153600} + O(\varepsilon^9). \quad (19.5)$$

19.2 The first finite-size functions

Let

$$f_2 = s^{-2} + H''(s), \quad f_3 = -2s^{-3} + H'''(s), \quad f_4 = 6s^{-4} + H''''(s). \quad (19.6)$$

Expansion of (18.6) to degree two gives

$$\begin{aligned} E_1(\varepsilon) &= \frac{1}{12} + \frac{sh'(s)}{12} - \frac{h''(s)/2 + h'(s)^2/4}{2f_2} + \frac{h'(s)f_3}{4f_2^2} \\ &\quad + \frac{f_4}{8f_2^2} - \frac{5f_3^2}{24f_2^3}. \end{aligned} \quad (19.7)$$

The $1/12$ is the Stirling correction, and $sh'(s)/12$ is the first Euler–Maclaurin correction. The remaining four terms are the vertical Gaussian saddle correction with amplitude $e^{h(s)/2}$. Although

r	Transition exponent	d_r in $\mathcal{D}(\varepsilon) = \sum d_r \varepsilon^{2r}$
1	5/2	$-1/72$
2	9/4	$-41/129600$
3	13/6	$-3749/342921600$
4	17/8	$-185201/411505920000$
5	21/10	$-3045727/149376648960000$
6	25/12	$-6551858030771/6679735362203904000000$

Table 4: Exact rational coefficients, not numerical fits. Each listed nonzero coefficient produces the transition in Theorem 14.3.

this display uses inverse powers of s , the expression has a removable singularity at zero. For stable evaluation use $\mathcal{Q} = s^2 f_2$, $\mathcal{Q}_3 = s^3 f_3$, and $\mathcal{Q}_4 = s^4 f_4$.

The first expansions are

$$E_0(\varepsilon) = \frac{\varepsilon^2}{48} + \frac{23\varepsilon^4}{51840} + \frac{331\varepsilon^6}{32659200} + O(\varepsilon^8), \quad (19.8)$$

$$E_1(\varepsilon) = \frac{\varepsilon^2}{36} + \frac{13\varepsilon^4}{5760} + \frac{30763\varepsilon^6}{195955200} + O(\varepsilon^8). \quad (19.9)$$

19.3 An independent check from the first central wave

The first nonconstant term in the centered product is $-t^2 \sum_{j \leq m} j^2/24$. Inverse Laplace extraction against $e^{Xt} t^{-m}$ therefore gives the relative X^{-2} correction

$$-\frac{(m-1)(m-2)}{24X^2} \sum_{j=1}^m j^2 = \frac{1}{X^2} \left(-\frac{m^5}{72} + \frac{m^4}{48} + \frac{m^3}{36} - \frac{m^2}{48} - \frac{m}{72} \right). \quad (19.10)$$

Its first three powers match exactly the ε^2 terms in $m\mathcal{D}, E_0, E_1/m$. This is a useful independent sign and normalization check. It is not used as a uniform approximation proof, since by itself it does not control all the other contour contributions.

[write] **Added 5 October 2026, batch 98.** The left side of (19.10) is the $j = 1$ term $a_1(m)(m-1)^2/M^2$ of Part I's expansion (3.8), with $a_1 = -S_2/24$ from (3.5) and $M = X$.

20 The crossover hierarchy and its infinitude

[write] **Added 5 October 2026, batch 98: Part I's question 2 answered.** Part I's research question 2 (§10) asks to extend the centered analysis into $m^{5/2} \lesssim N \ll m^3$, where the quadratic Bernoulli correction must itself be exponentiated, and to match the result to a small-parameter expansion of the Szekeres saddle, with a precise validity range and matching error. Theorem 14.1 does this with no gap: one expansion, with remainder $O_{a,J}(m^{-J-1})$, holds uniformly for every $N \geq am^2$, so it covers $N \ll m^3$, the scale $m^{5/2}$ and the compact- α Szekeres regime at once (§21). The exponentiated quadratic correction is the first entropy term $d_1 m \varepsilon^2 = -m^5/(72X^2)$; this section shows that $m^{5/2}$ is the exact threshold of the centered volume law and that it is the first of infinitely many transition scales. The question's lower end, $N/m^2 \rightarrow 0$, is not part of it and stays open (§25.3).

20.1 A uniform dilute expansion

For fixed r , analyticity at zero and Theorem 14.1 give, whenever $\varepsilon \rightarrow 0$,

$$\log \frac{p_m(N)}{\mathcal{V}_{r-1}(m, N)} = \mathbf{d}_r m \varepsilon^{2r} + \mathcal{O}_r(m \varepsilon^{2r+2} + \varepsilon^2 + m^{-1}). \quad (20.1)$$

In particular,

$$\log \frac{p_m(N)}{V_m(X)} = -\frac{m^5}{72X^2} - \frac{41m^9}{129600X^4} - \frac{3749m^{13}}{342921600X^6} + \cdots, \quad (20.2)$$

where the omitted endpoint and finite-size terms must be included when an actual numerical accuracy is specified. Equation (14.6), not the isolated display (20.2), is the full formula.

If $X \sim \lambda m^{2+1/(2r)}$, then $m \varepsilon^{2r} \rightarrow \lambda^{-2r}$ and every error in (20.1) tends to zero. This proves (14.11). More generally, within $X/m^2 \rightarrow \infty$, if $X \gg m^{2+1/(2r)}$ the ratio tends to one. If instead $X \ll m^{2+1/(2r)}$, then its logarithm is asymptotic to $\mathbf{d}_r m \varepsilon^{2r}$; the ratio tends to zero when $\mathbf{d}_r < 0$ and to infinity when $\mathbf{d}_r > 0$.

20.2 Necessity in the sharp centered boundary

By Lemma 17.1 and (19.3), $\mathcal{D}'(\varepsilon) < 0$ on $(0, 4)$, so $\mathcal{D}(\varepsilon) < 0$ there. Suppose first that $p_m(N)/V_m(X) \rightarrow 1$ with $N \geq am^2$. If ε did not tend to zero, a subsequence would remain in a compact subinterval bounded away from zero. The term $m\mathcal{D}(\varepsilon)$ would then tend to $-\infty$, while E_0 is bounded, a contradiction. Thus $\varepsilon \rightarrow 0$, and

$$\log \frac{p_m(N)}{V_m(X)} = -\frac{m \varepsilon^2}{72}(1 + \mathcal{O}(\varepsilon^2)) + \mathcal{O}(\varepsilon^2 + m^{-1}). \quad (20.3)$$

The logarithm can tend to zero only if $m \varepsilon^2 \rightarrow 0$. Conversely this condition makes the right side tend to zero. Since $m \varepsilon^2 = m^5/X^2$, this proves (14.7); the same display proves (14.8). This subsequence argument also covers oscillating choices of N rather than only power-law inputs.

20.3 Why the hierarchy does not terminate

It remains to prove that $\mathcal{D}$ is not a polynomial. For $s > 0$, direct integration of $h(s) = \log s - s/2 - \log(1 - e^{-s})$ yields

$$H(s) = \log s - 1 - \frac{s}{4} + \frac{\pi^2/6 - \text{Li}_2(e^{-s})}{s}. \quad (20.4)$$

Consequently, as $s \rightarrow \infty$,

$$B(s) = \frac{s}{4} + \frac{\pi^2}{6s} + \mathcal{O}(e^{-s}), \quad (20.5)$$

$$4 - \varepsilon(s) \sim \frac{8\pi^2}{3s^2}. \quad (20.6)$$

The entropy has the endpoint expansion

$$\mathcal{D}(\varepsilon) = \log 4 - 2 + \frac{\pi}{2\sqrt{6}}\sqrt{4 - \varepsilon} + \mathcal{O}(4 - \varepsilon), \quad \varepsilon \uparrow 4. \quad (20.7)$$

For example, it follows by substituting (20.5) and (20.4) into $\mathcal{D} = B - 1 - \log B + H$; the leading deviation is $\pi^2/(3s)$ and (20.6) gives the stated constant. Alternatively, (19.3) directly implies $\mathcal{D}'(\varepsilon) \rightarrow -\infty$.

If only finitely many coefficients $\mathbf{d}_r$ were nonzero, $\mathcal{D}$ would equal a polynomial near zero. Analytic continuation along the real interval $(0, 4)$ would make it that polynomial throughout, contradicting (20.7). Hence infinitely many $\mathbf{d}_r$ are nonzero, completing Theorem 14.3.

Remark 20.1 (An analytic endpoint is not a uniform partition regime). Equation (20.7) concerns the function $\mathcal{D}$ itself. The uniform theorem has a fixed lower bound $N \geq am^2$ and does not allow $a \downarrow 0$ with m . In particular the endpoint argument does not silently extend the partition expansion into the unrestricted Hardy–Ramanujan regime.

21 How the quadratic and cubic OEIS formulas fit together

21.1 The classical quadratic leading formula

Fix $\alpha > 0$ and take $N = \alpha m^2$ along integral values. Let s_α solve

$$\int_0^1 \frac{x}{e^{s_\alpha x} - 1} dx = \alpha. \quad (21.1)$$

Rearrangement of the leading terms of Theorem 14.1 gives

$$p_m(\alpha m^2) \sim \frac{s_\alpha}{2\pi m^2 \sqrt{2\alpha - (2\alpha + 1)e^{-s_\alpha}}} \exp(m\{2\alpha s_\alpha - \log(1 - e^{-s_\alpha})\}). \quad (21.2)$$

This is a parametrization of the classical Szekeres leading formula, not a new leading asymptotic [5]. For clarity about the endpoint constant, $X/m^2 = \alpha + 1/4 + 1/(4m)$. That last $1/(4m)$ contributes $e^{s_\alpha/4}$ to the leading amplitude. Together with $e^{h(s_\alpha)/2}$ it produces the factor in (21.2); dropping it gives an incorrect constant.

At $\alpha = 1$, formula (21.2) recovers the leading law for A206226 and the square subsequence of A097356. The previous repository report’s floor-phase and three-exception theorems are separate results and are not reproved or claimed here. Our theorem supplies a common expansion from the compact quadratic regime all the way through $\alpha \rightarrow \infty$.

[write] **Added 5 October 2026, batch 98.** Formula (21.2) is the leading constant of the report a097356-sqrt-restricted-partitions at $\alpha = 1$ (its constants u and C , labels `srp:eq:u-one` and `srp:eq:constants-one`; its u is s_1), and of Canfield’s account of Szekeres’ formula [5]: a second route, not a new result. The intake’s numerical check of (21.2) at $\alpha = 1$ against exact counts gives ratios 0.99223, 0.99611 and 0.99766 at $m = 48, 96, 160$, and $s_1 = 0.8146511367\dots$, equal to that report’s u .

21.2 Recovery of the uncentered cubic boundary

Let $N \sim cm^3$, $c > 0$. Then $\varepsilon \sim 1/(cm)$, so $m\mathcal{D}(\varepsilon) + E_0(\varepsilon) = o(1)$. On the other hand,

$$\log \frac{V_m(X)}{V_m(N)} = (m-1) \log \left(1 + \frac{m(m+1)}{4N} \right) \rightarrow \frac{1}{4c}. \quad (21.3)$$

Therefore

$$p_m(N) \sim e^{1/(4c)} \frac{N^{m-1}}{m! \Gamma(m)}. \quad (21.4)$$

This recovers the earlier repository and OEIS cubic law (Corollary 4.1 of Part I, and [2]); it is a compatibility check, not a claim that this law was missing.

For the exact A238608 slice $N = m^3$, Stirling expansion and (14.6) give

$$p_m(m^3) = \frac{e^{2m+1/4}m^{m-3}}{2\pi} \left(1 - \frac{61}{288m} + O(m^{-2})\right). \quad (21.5)$$

The leading term is posted in OEIS, and the correction is already in the inspected cubic report. The symbolic verifier checks that the new uniform formulation reproduces it exactly.

[write] **Added 5 October 2026, batch 98.** The correction $-61/(288m)$ is the first term of Part I's (4.11) and (4.12), and (21.4) is Corollary 4.1: both are second routes through Theorem 14.1, not new results. Part I gives the A238608 expansion through m^{-5} .

21.3 What the intermediate scales add

Between m^2 and m^3 , retaining only the shift $N \mapsto X$ is not enough. At $X \asymp m^{5/2}$ the first entropy term becomes visible. After incorporating it, the next transition lies at $m^{9/4}$; the next lies at $m^{13/6}$, and so on for the nonzero coefficients. At fixed N/m^2 , no finite number of these dilute entropy terms is a uniform substitute for the full analytic function $\mathcal{D}$. Thus the hierarchy explains both why many approximations work in a dilute range and why they eventually fail as the quadratic range is approached.

22 Analytic inversion and discrete thresholds

22.1 The leading inverse map

For Z, η_* in (14.12), write $X = Zx$. Equation (14.6) becomes

$$0 = (m-1)\log x + m\mathcal{D}(\eta_*/x) + E_0(\eta_*/x) + \frac{E_1(\eta_*/x)}{m} + \dots \quad (22.1)$$

Dividing by m , the leading equation is $\log x + \mathcal{D}(\eta_*/x) = 0$. Setting $\varepsilon = \eta_*/x$ gives exactly (14.13). By (19.3),

$$\frac{d}{d\varepsilon}(\varepsilon e^{-\mathcal{D}(\varepsilon)}) = e^{-\mathcal{D}(\varepsilon)}(1 - \varepsilon\mathcal{D}'(\varepsilon)) = e^{-\mathcal{D}(\varepsilon)}B(\varepsilon) > 0. \quad (22.2)$$

At zero this derivative is one; as $\varepsilon \uparrow 4$, its image tends to $4e^{-(\log 4 - 2)} = e^2$. This proves the existence and analyticity of the inverse chart on $[0, e^2)$.

22.2 First and all higher corrections

Put $x = \mathcal{R}(\eta_*)e^{\omega/m + O(m^{-2})}$ in (22.1). Its constant-order terms are

$$(1 - \varepsilon\mathcal{D}'(\varepsilon))\omega - \log \mathcal{R} + E_0(\varepsilon) = 0, \quad (22.3)$$

which proves (14.14) and (14.15). More generally, insert

$$X = Z\mathcal{R}(\eta_*) \exp\left(\sum_{k=1}^K \frac{\omega_k(\eta_*)}{m^k} + O_{a,K}(m^{-K-1})\right). \quad (22.4)$$

At each new order the unknown ω_k enters linearly with coefficient $1 - \varepsilon \mathcal{D}'(\varepsilon) = B(s)$, bounded away from zero. The remaining terms are already known analytic functions. This recursion, or the analytic implicit-function theorem applied separately to each finite truncation with parameter $1/m$, proves all fixed orders uniformly on the compact parameter ranges of the forward theorem. The finite forward remainder transfers through the same nonvanishing derivative. This completes Theorem 14.4.

Remark 22.1 ([write] The all-orders inverse in detail; added 5 October 2026, batch 98). The two sentences above are the whole argument for “every fixed order”. Written out: put $\hbar_1 = 1/m$ and

$$F_K(x; \eta_*, \hbar_1) = (1 - \hbar_1) \log x + \mathcal{D}(\eta_*/x) + \sum_{j=0}^K \hbar_1^{j+1} E_j(\eta_*/x).$$

Dividing (14.6) (with $J = K$) by m shows that the exact $x = X/Z$ satisfies $F_K(x; \eta_*, 1/m) = O_{a,K}(m^{-K-2})$. At $\hbar_1 = 0$, $F_0(x; \eta_*, 0) = \log x + \mathcal{D}(\eta_*/x)$ is strictly increasing in x with derivative $B(s)/x > 0$ ($\varepsilon = \eta_*/x$), by (19.3), and vanishes at $x = \mathcal{R}(\eta_*)$; since $\eta_* = \varepsilon x = \varepsilon e^{-\mathcal{D}(\varepsilon) + O(1/m)}$ by the forward theorem, η_* lies in a compact subset of $[0, e^2)$, and, as $B > 1$, the case $K = 0$ already places the exact x within $O(1/m)$ of $\mathcal{R}(\eta_*)$. (Dated note, 5 October 2026: the first version read “vanishes at $x = \mathcal{R}(\eta_*)$; so the case $K = 0$ already places...”, stating the compactness of η_* only in the next sentence; the order was corrected after the independent check recorded at the head of §25.) In the regime $N \geq am^2$ one has $\varepsilon \leq \varepsilon_a$, hence η_* stays in a compact subset of $[0, e^2)$, on which F_K is analytic near $(\mathcal{R}(\eta_*), \eta_*, 0)$ and $\partial_x F_K = B(s)/x$ there is bounded away from zero. The analytic implicit-function theorem gives a root $x_K(\eta_*, \hbar_1)$ analytic in $\hbar_1$ near 0, uniformly in η_* , with $\log x_K = \log \mathcal{R} + \sum_{k \leq K} \omega_k \hbar_1^k + O(\hbar_1^{K+1})$, where $\omega_1 = \omega$ of (14.14) and each ω_k solves a linear equation with coefficient $B(s)$, as stated. The mean-value theorem in x , with $|\partial_x F_K|$ bounded below, turns $F_K(x; \eta_*, 1/m) = O(m^{-K-2})$ into $\log x - \log x_K = O(m^{-K-2})$. Together these give (22.4), uniformly on the regime.

The first dilute inverse coefficients are

$$\mathcal{R}(\eta_*) = 1 + \frac{\eta_*^2}{72} + \frac{7\eta_*^4}{259200} + \frac{347\eta_*^6}{2743372800} + O(\eta_*^8), \quad (22.5)$$

$$\omega(\eta_*) = -\frac{\eta_*^2}{144} + \frac{67\eta_*^4}{259200} + \frac{653\eta_*^6}{1371686400} + O(\eta_*^8). \quad (22.6)$$

For example the inverse parameter itself is

$$\varepsilon = \eta_* - \frac{\eta_*^3}{72} + \frac{43\eta_*^5}{259200} - \frac{5639\eta_*^7}{2743372800} + O(\eta_*^9). \quad (22.7)$$

These coefficients were generated by exact Lagrange inversion and checked by substitution into (14.13).

22.3 A safe integer-threshold statement

Let $T_m(Y) = \min\{N \geq 0 : p_m(N) \geq Y\}$. The injection that adds a part of size one makes p_m nondecreasing, but does not provide a canonical real interpolation. Define the smooth logarithmic surrogate

$$\mathcal{L}_J(X) = \log V_m(X) + m\mathcal{D}(m^2/X) + E_0(m^2/X) + \sum_{j=1}^J \frac{E_j(m^2/X)}{m^j}. \quad (22.8)$$

On any admissible compact parameter range,

$$\mathcal{L}'_J(X) = \frac{mB(s) - 1 + O_a(1)}{X} \asymp_a \frac{m}{X} > 0 \quad (22.9)$$

for sufficiently large m . Its leading derivative follows directly from (19.3); the derivatives of the remaining analytic functions are bounded.

Proposition 22.2 (Threshold error transfer). *Suppose the true threshold and a fixed relative neighborhood of it lie in $N \geq am^2$. Let X_J solve $\mathcal{L}_J(X_J) = \log Y$. Then*

$$T_m(Y) = X_J - \frac{m(m+1)}{4} + O_{a,J} \left(\frac{X_J}{m^{J+2}} + 1 \right). \quad (22.10)$$

Proof. Theorem 14.1 bounds the logarithmic error at every integer sample by Cm^{-J-1} . By (22.9), moving the smooth root by $C'X_Jm^{-J-2}$ changes the surrogate logarithm by at least that amount, with either sign, if C' is large enough. Round the two comparison locations outward to integers, adding a bounded displacement. The lower one has count below Y and the upper one has count at least Y . Monotonicity supplies the bracket. $\square$

A symbolic asymptotic expansion alone does not furnish an explicit numerical value of the error constant or an exact ceiling formula. Even a very small relative error can correspond to a large absolute index error when X is enormous.

[write] **Added 5 October 2026, batch 98: an instance of repository results.** For fixed $m \geq 2$ the counts $p_m(N)$ are strictly increasing in $N \geq 1$ (the difference $p_m(N+1) - p_m(N)$ counts partitions of $N+1$ into parts from $\{2, \dots, m\}$), so $T_m(Y)$ is the integer staircase of `p0:thm:staircase` in the transseries-and-inversion volume [10], and the proof above is that theorem's bracketing argument (its parts (1)–(2)) with an explicit chart error. As in Part I's dated notes in §6, no novelty is claimed for the method.

22.4 Lambert- W inversion along a power-law slice

A complementary inverse lets m vary on a fixed centered slice $X = \lambda m^\gamma$, $\gamma > 2$, $\lambda > 0$. Ignoring bounded integer rounding for the moment, Stirling gives

$$\begin{aligned} \log p_m(N) &= (\gamma - 2)m \log m + (2 + \log \lambda)m - \gamma \log m - \log(2\pi\lambda) \\ &\quad + m\mathcal{D}(\lambda^{-1}m^{2-\gamma}) + E_0(\lambda^{-1}m^{2-\gamma}) + \frac{E_1(\lambda^{-1}m^{2-\gamma}) - 1/6}{m} + O(m^{-2}). \end{aligned} \quad (22.11)$$

For $L = \log Y$, $\beta = \gamma - 2$ and $b_\lambda = 2 + \log \lambda$, the leading inverse is

$$m_0 = \frac{L}{\beta W(e^{b_\lambda/\beta} L/\beta)}. \quad (22.12)$$

Successive corrections are obtained by substituting $m = m_0 + \Delta$ into (22.11); the leading derivative is $\beta \log m_0 + \beta + b_\lambda$. All the retained entropy sectors must be included in that substitution at their actual sizes. This is an application of the repository's transseries-inversion program, not a new inversion method. Bounded rounding of N alters the continuous profile logarithm by $O(m/X)$ and does not affect its leading asymptotic; exact thresholds still require an argument such as Proposition 22.2.

[write] **Added 5 October 2026, batch 98: an instance of repository results.** The “repository's transseries-inversion program” is the transseries-and-inversion volume [10]: the core $\beta m \log m + b_\lambda m = L$,

solved by the principal branch of W in (22.12), is the Lambert core of the inverse gamma function in `p6:thm:gamma`, and the substitution $m = m_0 + \Delta$ is its perturbed inversion `p0:thm:perturbed-inversion`. At $\gamma = 3$ one has $\beta = 1$ and $m_0 = L/W(\lambda e^2 L)$, which is Part I's cubic core (6.2) with c replaced by λ (and $X \sim N$ on that slice, so $\lambda \sim c$).

23 What kind of transseries has been obtained?

The answer contains two different expansions. First, $\mathcal{D}(\varepsilon)$ is a genuinely convergent analytic germ in ε , with an effective rational coefficient formula. Second, (14.6) is a fixed-order asymptotic expansion in $1/m$, uniform in a compact continuum parameter. We have not proved convergence, Gevrey bounds, Borel summability or optimal truncation for the latter.

On a power-law slice $X = \lambda m^\gamma$, its entropy factor resolves ordered exponential corrections

$$\exp \left(\sum_{r \geq 1} d_r \lambda^{-2r} m^{1-2r(\gamma-2)} \right). \quad (23.1)$$

Only finitely many exponents in (23.1) are nonnegative when $\gamma > 2$. After those terms are retained in the exponent, the others can be expanded to any desired fixed algebraic order, together with the endpoint and $1/m$ functions. At a transition exponent the corresponding term is a nontrivial constant rather than a decaying correction. This is the precise sense in which the report supplies an exponential asymptotic hierarchy useful for transseries calculations.

It would be too strong to call this a complete resurgent transseries. The exponentially small contributions of other roots of unity are not individually identified; their actions, amplitudes and possible Stokes behavior are not derived. The minor-arc estimates provide the uniform algebraic accuracy needed here, not a classification of every beyond-all-orders sector.

[write] **Added 5 October 2026, batch 98.** Part I states the same limitation for its centered theorem (the remark after the proof of Theorem 3.1) and poses the individual waves as its question 1. Part I's dilute expansion (5.8) does resolve genuine q^{-m} sectors, but only for $N \geq m^3/\eta_0$ (Theorem 5.1); nothing in Part II resolves exponentially small sectors at the quadratic scale.

24 Computational verification and reproducibility

The package contains independent exact and numerical checks. Exact arithmetic is used for the partition counts and rational symbolic identities. The evaluation of special functions and saddle roots uses 55-digit `mpmath` arithmetic and is *not* interval-certified.

24.1 Exact computations

The dynamic program uses (15.1). As an independent recurrence, logarithmic differentiation of (13.1) gives

$$N p_m(N) = \sum_{k=1}^N \sigma_m(k) p_m(N-k), \quad \sigma_m(k) = \sum_{\substack{d|k \\ 1 \leq d \leq m}} d. \quad (24.1)$$

The verifier compares these methods through $N = 400$ for $m = 2, 3, 5, 8, 12$, including exact divisibility of each recurrence numerator. It also reproduces the 17 displayed A238608 values

at $m = 0, \dots, 16$ from the retrieved OEIS entry [2]. In total it executes 4,017 exact arithmetic assertions.

The symbolic script derives the six d_r , the displayed E_0, E_1 series and the inverse coefficients. It checks four identities: the implicit saddle equation, an independent entropy substitution, the inverse substitution, and the known cubic correction $-61/288$. The first-wave calculation (19.10) supplies a further transparent mathematical cross-check of the normalization.

24.2 Uniform-profile diagnostics

The numerical run stores 46 exact partition samples with $m \leq 96$ and $N \leq 268566$. It includes $N/m^2 = 1/4, 1, 4$, centered $m^{5/2}$ and $m^{9/4}$ slices, and cubic samples. Table 5 displays signed logarithmic errors for $N = m^2$. Here $\mathcal{L}_0 = \log V + m\mathcal{D} + E_0$ and $\mathcal{L}_1 = \mathcal{L}_0 + E_1/m$.

m	$\log p_m(m^2) - \mathcal{L}_0$	$\log p_m(m^2) - \mathcal{L}_1$	$m^2(\log p - \mathcal{L}_1)$
16	1.0858192×10^{-3}	$-5.5491004 \times 10^{-5}$	-0.01420570
24	7.4303668×10^{-4}	$-2.4472407 \times 10^{-5}$	-0.01409611
32	5.6443633×10^{-4}	$-1.3708705 \times 10^{-5}$	-0.01403771
48	3.8104962×10^{-4}	$-6.0670539 \times 10^{-6}$	-0.01397849
64	2.8756776×10^{-4}	$-3.4054133 \times 10^{-6}$	-0.01394857
96	1.9289706×10^{-4}	$-1.5102484 \times 10^{-6}$	-0.01391845

Table 5: Exact-count comparisons with the continuum expansion. The scaled errors support, but do not prove, the $O(m^{-2})$ remainder.

For the separate exact-saddle formula, the first Edgeworth correction reduces the signed logarithmic error at $m = 96, N = 9216$ from about -1.06104×10^{-3} to 4.92792×10^{-7} . Thus the finite-sum and continuum implementations provide distinct numerical tests of the analysis.

24.3 Crossover and inverse diagnostics

At the first centered transition with $\lambda = 1/2$, the exact ratios $p_m(N)/V_m(X)$ are approximately 0.951023 at $m = 16$, 0.947144 at $m = 64$, and 0.946743 at $m = 96$. Their proved limit is $e^{-1/18} \approx 0.945959$. The small bounded rounding used to select integer N is recorded in the CSV data rather than suppressed.

There is an instructive limitation in the second-transition samples. At $X \approx (3/4)m^{9/4}$ the ratio after removing only the first entropy term is still greater than one for every sampled $m \leq 96$, although its proved limiting constant is below one. The positive E_0 correction is still larger than the tiny second entropy effect at these sizes. We retain this evidence rather than presenting the finite data as proof that the limiting sign has already appeared.

[write] **Added 5 October 2026, batch 98: where the sign should change.** Evaluating $m(\mathcal{D}(\varepsilon) - d_1\varepsilon^2) + E_0(\varepsilon) + E_1(\varepsilon)/m$ from the series (19.8), (19.9) and d_2, d_3, d_4 on the slice $X = \frac{3}{4}m^{9/4}$ gives 0.0028418 at $m = 96, N = 19308$, against the exact $\log(p_m(N)/\mathcal{V}_1(m, N)) = 0.0028414$ recorded in `data/02-crossover-crossover_checks.csv`. The same truncated expression changes sign between $m = 1373$ and $m = 1374$ and tends to $d_2/(3/4)^4 = -0.0009998\dots$. This locates the expected crossing; it is an estimate from a truncated expansion, with no bound on the omitted terms at that size, and is recorded as §25.9. *Dated note (5 October 2026)*. The first version of this note read “gives 0.0028419 at $m = 96, \dots$ (with $N = 19308$)”; 0.0028419 (0.002841865) is the value of the expression at the real $X = \frac{3}{4} \cdot 96^{9/4}$, while at the integer $N = 19308$, to which the exact value refers, it is 0.0028418 (0.002841819). The independent

check recorded at the head of §25 found the sign change at $m = 1373/1374$ correct (values $+3.0 \cdot 10^{-8}$ and $-3.35 \cdot 10^{-7}$; the sign changes there both for the real X and for the integer N) and its framing as an estimate appropriate.

For $Y = p_m(m^2)$, the first corrected inverse chart has relative centered- X errors 8.92656×10^{-5} at $m = 16$, 5.58154×10^{-6} at $m = 64$, and 2.48072×10^{-6} at $m = 96$. Multiplication by m^2 gives approximately 0.022852, 0.022862, and 0.022862, consistent with (14.15). These are sample-value diagnostics, not certificates for rounding arbitrary thresholds.

24.4 Package contents and commands

The source is self-contained LaTeX with an internal bibliography. The included files are:

File	Role
<code>article.tex</code> , <code>article.pdf</code>	This manuscript and its compiled PDF.
<code>code/asymptotics.py</code>	Stable forward, finite-saddle and inverse numerical evaluators.
<code>code/derive_coefficients.py</code>	Exact rational coefficients and symbolic assertions.
<code>code/verify.py</code>	Exact recurrences and 55-digit diagnostic comparisons.
<code>data/</code>	Exact samples, coefficient JSON, diagnostic CSVs and execution records.
<code>README.md</code> , <code>SOURCE_AUDIT.md</code>	Build instructions, provenance and novelty limits.

From the package root, run

```
python -m pip install -r requirements.txt
python code/derive_coefficients.py
python code/verify.py
pdflatex -interaction=nonstopmode -halt-on-error article.tex
pdflatex -interaction=nonstopmode -halt-on-error article.tex
```

The recorded environment is Python 3.13.5 and mpmath 1.3.0. Runtime and tool versions are evidence about the performed run, not assumptions of the mathematical theorems. No Lean or Rocq verification was performed, and the numerical calculations do not instantiate the constants in the uniform remainder proofs.

[write] **Added 5 October 2026, batch 98: the shipped layout.** In this report every delivered file shipped carries the prefix `02-crossover-`: the three programs `asymptotics.py`, `derive_coefficients.py` and `verify.py` and the delivered `build.sh` are in `code/` (the build script compiles `article.tex`, which is now this report); the ten records, including the requirements file, are in `data/`; the two audits `SOURCE_AUDIT.md` and `QA.md` are at the report root. Not shipped: the manuscript's `article.tex` and `article.pdf` (Part II is its text), its `README.md`, `SHA256SUMS.txt` (verified, 20 of 20) and `data/symbolic_run.txt`, a byte copy of the shipped coefficient file. `verify.py` imports `asymptotics` and reads `data/coefficients.json` by their delivered names, and the programs write unprefixed files into a `data/` directory beside `code/`, so they must be rerun on a copy with the delivered names restored, never in place; the report `README` gives the commands. On 5 October 2026 such a rerun passed here: `verify.py` in 80 s (the record says 12.8 s), with the five CSV files byte-identical, `exact_samples.json` identical up to line endings, and `verification_summary.json` and the standard output (against `verification_run.txt`) identical apart from `elapsed_seconds` and line endings; `derive_coefficients.py` in 14 s, reproducing `coefficients.json` up to line endings. The 17 A238608 terms embedded in `verify.py` are OEIS data (CC BY-SA 4.0).

25 Further questions and research

The following questions are precise continuations, not results of this report.

[write] **Added 5 October 2026, batch 98: what this section holds.** Questions 25.1–25.7 are manuscript 09’s own (its §§13.1–13.7), printed as delivered with dated notes. Questions 25.8 and 25.9 were moved here in the write, under the repository’s standing rule that unproved claims are kept as questions: the first collects the non-claims of manuscript 09’s §23 and of its delivery README, the second its finite-size observation in §24. Where a question overlaps one of Part I’s (§10) or one of the report `a097356-sqrt-restricted-partitions`, the overlap is named and both sources are credited; the questions are printed once, here. Apart from the two condensed proofs completed in Remarks 18.1 and 22.1, every statement manuscript 09 asserts is proved in its text, and the write found none of its claims to be false.

Dated note (5 October 2026). An independent adversarial check of the results added at this write, made by the intake after the write, found them valid: all six steps of Remark 18.1, Remark 22.1, and the dated note in §24 with question 25.9, which present the sign change at $m = 1373/1374$ correctly as an estimate. The check was a careful reading with independent computations: its own implementation of the operator (18.7) (its E_1 agrees with (19.7) to 12 digits); exact $p_m(N)$ by the recurrence for $m = 10, \dots, 240$ and N/m^2 from 0.05 to 10^4 ($N \leq 3 \cdot 10^6$), whose residuals after $J = 0, 1, 2$ corrections, scaled by m^{J+1} , approach E_{J+1} uniformly in N/m^2 ; an independent second inverse coefficient ω_2 , with next remainder $O(m^{-3})$; measured arc ratios for step (iv) of Remark 18.1; and a recomputation of the truncated expression on the slice. It is not a formal verification or an external review. It led to three changes, each recorded with a dated note where it was made: the value at $m = 96$ in the note of §24 (0.0028418 at $N = 19308$, not 0.0028419); in Remark 22.1, the compactness of η_* in $[0, e^2]$ is now stated before the step $K = 0$, which then gives $|\log x - \log \mathcal{R}(\eta_*)| = O(1/m)$; and numerical evidence in question 25.9.

25.1 Are all entropy coefficients strictly negative?

The first six d_r are negative and infinitely many are nonzero. Is $[z^{2r}]B(z)^{2r+1} > 0$ for every $r \geq 1$? A positive integral or combinatorial representation of these diagonal coefficients would settle the question and make every exponent $2 + 1/(2r)$ an unconditional member of the hierarchy. The alternating Bernoulli coefficients of B mean that coefficientwise positivity is not immediate.

[write] **Added 5 October 2026, batch 98: evidence.** In exact rational arithmetic, by two independent routes (the Lagrange formula (14.9), and the series of $\mathcal{D} = s/\varepsilon - 1 - \log(s/\varepsilon) + H(s)$ after solving $s = \varepsilon B(s)$ by fixed-point iteration), $d_1, \dots, d_{16}$ all agree and are all negative; $d_1, \dots, d_6$ are those of Table 4. This is evidence, not a proof; what is missing is an argument valid for every r . An answer to question 25.2 would give negativity for all large r if the endpoint singularities dominate.

25.2 What controls the large-order behavior of d_r ?

The real endpoint at $\varepsilon = 4$ has a square-root singularity. Does it dominate the Taylor coefficients, or do nearer complex critical values of $s/B(s)$ intervene? Determine the radius of convergence, nearest singularities, and the leading large- r law for d_r . The endpoint analysis alone does not answer this complex-analytic question.

[write] **Added 5 October 2026, batch 98: a conjectural law and its evidence.** If $\varepsilon = \pm 4$ were the only singularities on the circle of convergence, with the local form (20.7) at both ($\mathcal{D}$ is even), the standard transfer from $\sqrt{1 - \varepsilon/4}$ would give

$$d_r \sim -\sqrt{\pi/6} (2r)^{-3/2} 16^{-r} \quad (r \rightarrow \infty),$$

and hence negativity of all large d_r . The exact values agree with this law to the ratios 0.869, 0.955, 0.987 and 0.996 at $r = 1, 7, 12$ and 16, increasing monotonically in between. This is numerical evidence only. Two

things are missing: a proof that $s/B(s)$ has no complex critical value of modulus at most 4 other than at the real ends, and a justification of the transfer, since the expansion of $\mathcal{D}$ at $\varepsilon = 4$ in powers of $\sqrt{4 - \varepsilon}$ carries corrections of size e^{-s} with $s \asymp (4 - \varepsilon)^{-1/2}$ (from (20.4)), which are not analytic in $\sqrt{4 - \varepsilon}$ at the endpoint.

25.3 Can the opposite moving-saddle endpoint be made uniform?

Extend the expansion as $N/m^2 \rightarrow 0$, allowing $s \rightarrow \infty$. Here $\mathcal{Q}(s) \rightarrow 0$, so the present compact-curvature proof is insufficient. A successful result should identify its maximal range and match to unrestricted-partition asymptotics rather than merely substituting a large s into a compact-parameter theorem.

[write] **Added 5 October 2026, batch 98.** This is the end $\alpha \rightarrow 0$ of the question “Uniform limits as the saddle moves to zero or infinity” (Section 14.6) of the report `a097356-sqrt-restricted-partitions`; Theorem 14.1 answers its end $\alpha \rightarrow \infty$. Part I’s question 2 (§10) does not include it.

25.4 Can individual exponential waves be resolved?

Refine the collective minor-arc estimate into explicit contributions from the dominant nontrivial roots of unity. Determine the parameter ranges in which each contribution is visible, and how those ranges interact with the entropy hierarchy. This would move beyond fixed-order Poincaré expansions toward a genuine multi-sector transseries.

[write] **Added 5 October 2026, batch 98.** The same question at the cubic scale is Part I’s question 1 (“Optimal truncation and individual waves”, first target the $q = -1$ wave), and at the quadratic scale it is Section 14.3 (“Exponential sectors and optimal truncation”) of the report `a097356-sqrt-restricted-partitions`. One uniform answer for $N \geq am^2$ would settle all three.

25.5 Can the error constants support exact inversion?

Replace the existence of constants in the uniform expansion by computable interval bounds, then combine them with Proposition 22.2 to certify exact thresholds for large Y . The difficult issue is not obtaining many numerical digits of a saddle, but controlling all analytic remainders strongly enough to exclude neighboring integers.

[write] **Added 5 October 2026, batch 98.** This is Part I’s question 8 (“Certified inverse thresholds”), extended from $N \geq c_0 m^3$ to $N \geq am^2$; the separation condition of `p0:thm:staircase(2)` is the test such an interval must pass.

25.6 How general is the centering hierarchy?

For restricted products $\prod_{j \leq m} (1 - q^{w_j})^{-1}$, centering by $\frac{1}{2} \sum w_j$ again removes the linear term. Under regularity and aperiodicity assumptions on the weights, do analogues of $\mathcal{D}, E_j$ and an infinite hierarchy follow? Any extension must preserve a lattice span-one argument or explicitly account for residue classes; the reserved $j = 1$ factor used here cannot simply be assumed available.

[write] **Added 5 October 2026, batch 98.** This is Part I’s question 7 (“Other weight systems”), which also asks to separate a universal volume and cumulant law from periodic obstructions. The weights w_j here are unrelated to the coordinate ζ of §18.

25.7 Can the analytic proof be formalized efficiently?

A realistic route begins with the exact product, positivity and recurrences, then formalizes the two contour modulus lemmas, uniform Euler–Maclaurin bounds on a compact family, and the finite Gaussian coefficient operator. The formal challenge is managing quantitative uniformity and complex analytic remainders, not just verifying the listed rational coefficients.

[write] **Added 5 October 2026, batch 98.** Part I’s formalization route (its Section 9) and its question 9 (“A proof-producing asymptotic pipeline”) propose the same layering for $N \geq c_0 m^3$. Its starting point, the Lean file `PartitionBoundedParts.lean` (§1.3), formalizes the finite-product generating function only; no statement of Part II is formalized.

25.8 Convergence, Borel summability and optimal truncation of the $1/m$ series

Moved here in the write (5 October 2026, batch 98), from manuscript 09’s §23, its notation paragraph in §13 and its delivery README. The expansion (14.6) is proved at every fixed order J , with constants depending on J . Manuscript 09 does not prove, and does not claim, convergence of $\sum_j E_j(\varepsilon) m^{-j}$, Gevrey bounds on the E_j , Borel summability, an optimal truncation order growing with m , or any uniformity in a growing J ; in its words, calling the result a complete resurgent transseries “would be too strong”. Determine the growth of $\sup_{0 \leq \varepsilon \leq \varepsilon_a} |E_j(\varepsilon)|$ in j (the operator (18.7) makes the E_j computable), the optimal truncation and the size of the resulting exponentially small error, and how that error compares with the root-of-unity contributions of question 25.4. Part I’s question 1 asks the same for its cubic expansion.

25.9 When does the second-transition sign appear?

Moved here in the write (5 October 2026, batch 98), from manuscript 09’s §24. On the slice $X \approx \frac{3}{4} m^{9/4}$ the ratio $p_m(N)/\mathcal{V}_1(m, N)$ exceeds one for every sampled $m \leq 96$, although its proved limit $\exp(d_2/(3/4)^4)$ is below one; manuscript 09 attributes this to the positive term E_0 . The truncated expansion puts the change of sign between $m = 1373$ and $m = 1374$ (the dated note in §24). Missing: a proof, which needs explicit remainder constants in Theorem 14.1 at that size (question 25.5), or an exact count at $m \approx 1.4 \cdot 10^3$, $N \approx 8.1 \cdot 10^6$, where the dynamic program of §24 needs about 10^{10} big-integer additions. *Dated note (5 October 2026): numerical evidence only.* The independent check recorded at the head of §25 computed $\log p_m(N)$ by floating-point quadrature of the Cauchy integral on the exact-saddle circle. Against exact counts it gives 0.0028414007 at $m = 96$, $N = 19308$ (the recorded value) and agrees to $2 \cdot 10^{-10}$ at $m = 20$, $N = 400$; the omitted minor arcs are of size e^{-18} there and e^{-530} at $m = 1373$. On the integer slice $N = \text{round}(\frac{3}{4} m^{9/4} - m(m+1)/4)$ the actual $\log(p_m(N)/\mathcal{V}_1(m, N))$ is $+2.96 \cdot 10^{-8}$ at $m = 1373$ and $-3.35 \cdot 10^{-7}$ at $m = 1374$, so the actual counts also appear to change sign there. The computation takes seconds, but it is floating point and uncertified: it supports the estimate and replaces neither a proof nor an exact count.

26 Conclusion of Part II

The restricted-partition family admits one centered, fixed-order expansion throughout $N \geq am^2$ for each fixed $a > 0$, without an upper bound on N . Its zero-saddle endpoint remains regular after a scaled local coordinate is used and one factor is reserved in the outer-arc bound. The result connects

the repository’s quadratic and cubic calculi and identifies a hierarchy of intermediate approximation scales.

The first scale is $m^{5/2}$, distinct from the uncentered cubic boundary. Higher scales follow from explicit rational entropy coefficients, and the hierarchy is proved not to terminate. The same analytic entropy gives an invertible forward-to-inverse parameter map, with all fixed-order corrections and a separate threshold error rule. These are the proved conclusions of the report. Universal coefficient negativity, complete exponential sectors, the opposite endpoint, certified numerical thresholds and global priority remain outside its established scope.

A A compact all-order coefficient recipe

[write] **Added 5 October 2026, batch 98.** This appendix belongs to Part I.

For a requested order K , the following is a finite exact procedure. Compute $S_{2r}(m)$ for $1 \leq r \leq K$ by Faulhaber’s formula or finite symbolic summation. Form γ_r , then a_j from (3.4). Let $x = 1/m$, $h = (x + x^2)/(4c)$, and form, modulo x^{K+1} ,

$$Q(x, c) = \sum_{j=0}^K a_j(1/x) \frac{x^{4j}}{c^{2j}} \prod_{i=1}^{2j} (1 - ix) (1 + h)^{-2j}.$$

Although the expression uses $a_j(1/x)$, the combined summand has no negative powers: its lowest possible degree is j . Then

$$\begin{aligned} \sum_{r=1}^K L_r(c) x^r &\equiv \log Q(x, c) + (x^{-1} - 1) \log(1 + h) - \frac{1}{4c} \\ &\quad - 2 \sum_{r \geq 1} \frac{B_{2r}}{2r(2r-1)} x^{2r-1} \pmod{x^{K+1}}. \end{aligned}$$

Only finitely many Stirling terms contribute. Apply (4.4) to obtain relative coefficients. This is the algorithm implemented in `derive_coefficients.py`; it uses exact rational arithmetic. For dilute coefficients, use (5.3), (5.2), and (5.5) instead. No numerical sequence values are needed to discover any of the displayed coefficients.

References

- [1] OEIS Foundation Inc., *A238016: Number of partitions of n^k into parts that are at most n* . Entry by Alois P. Heinz; asymptotic comments by V. Kotešovec (2015). Retrieved 1 October 2026. <https://oeis.org/A238016>.
- [2] OEIS Foundation Inc., *A238608: Number of partitions of n^3 into parts that are at most n* . Entry by Alois P. Heinz; leading asymptotic and its cn^3 extension by V. Kotešovec (2015). Retrieved 1 October 2026. <https://oeis.org/A238608>.
- [3] OEIS Foundation Inc., *A238010: Number of partitions of k^n into parts that are at most n* . Entry by Alois P. Heinz; asymptotic comment by V. Kotešovec (2015). Retrieved 1 October 2026. <https://oeis.org/A238010>.
- [4] OEIS Foundation Inc., *A258670: Number of partitions of $(2n)!$ into parts that are at most n* . Retrieved 1 October 2026. <https://oeis.org/A258670>.

- [5] E. Rodney Canfield, *From recursions to asymptotics: On Szekeres' formula for the number of partitions*, Electronic Journal of Combinatorics **4**, no. 2 (1997), Research Paper 6, 16 pp. <https://doi.org/10.37236/1321>.
- [6] Karl Dilcher and Christophe Vignat, *An explicit form of the polynomial part of a restricted partition function*, Research in Number Theory **3** (2017), article 1. <https://doi.org/10.1007/s40993-016-0065-3>.
- [7] Andrew V. Sills and Doron Zeilberger, *Formulae for the number of partitions of n into at most m parts (using the quasi-polynomial ansatz)*, arXiv:1108.4391 (2011). <https://arxiv.org/abs/1108.4391>.
- [8] Cormac O'Sullivan, *Partitions and Sylvester waves*, arXiv:1702.03611 (2017). <https://arxiv.org/abs/1702.03611>.
- [9] NIST Digital Library of Mathematical Functions, §5.11, *Asymptotic expansions of the gamma function*, especially 5.11.1 and the remainder bounds in §5.11(ii). Retrieved 1 October 2026. <https://dlmf.nist.gov/5.11>.
- [10] ProveIt, *Transseries: the polynomial-logarithmic calculus, series reversal at infinity, and inversion*, repository volume `Analysis/Transseries/docs/series-and-transseries/Transseries_And_Inversion/transseries_and_inversion.tex`, theorems labelled `p0:thm:staircase`, `p0:thm:perturbed-inversion` and `p6:thm:gamma`. Added in writing (batch 74).
- [11] ProveIt research-report collection, `SetTheory/Cardinals/docs/reports/generating-functions-and-asymptotics/oeis-sequence-asymptotics/a097356-sqrt-restricted-partitions`, research question “Uniform limits as the saddle moves to zero or infinity”. Unrefereed and not formalized. Added in writing (batch 74).
- [12] Vladimir Reshetnikov, *ProveIt repository*, inspected module `PartitionBoundedParts.lean` (full path in Section 1.3), main branch, retrieved 1 October 2026. <https://github.com/VladimirReshetnikov/ProveIt>.
- [13] OEIS Foundation Inc., *A097356: Number of partitions of n into parts not greater than $\sqrt{n}$* . Entry by Reinhard Zumkeller; asymptotic comments by V. Kotešovec. Inspected 4 October 2026 by manuscript 09 of batch 98. <https://oeis.org/A097356>. Added in writing (batch 98).